



TRƯỜNG ĐẠI HỌC
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THIẾT KẾ SOC

SOC Design
SOCD334264

Chap 1 INTRODUCTION

Microelectronics trends

MICROELECTRONICS TRENDS



Better device technology



reduced in device sizes

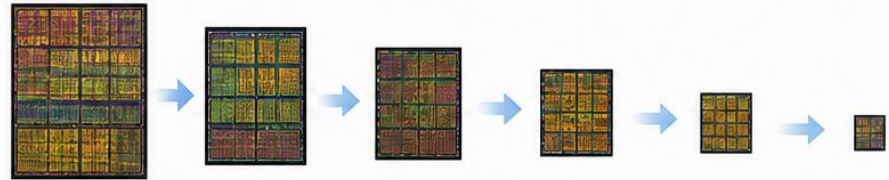


more on chip devices
> higher density

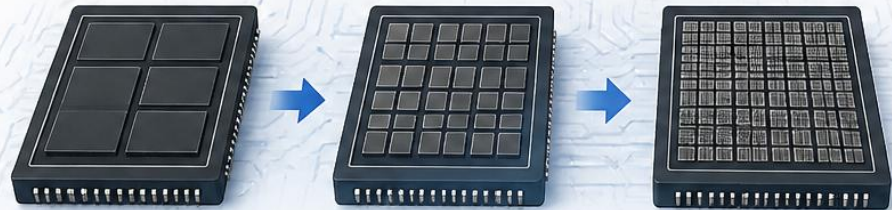


higher performances

180nm ~2000 90nm ~2005 45nm ~2010 22nm ~2015 7nm ~2018 3nm ~2022



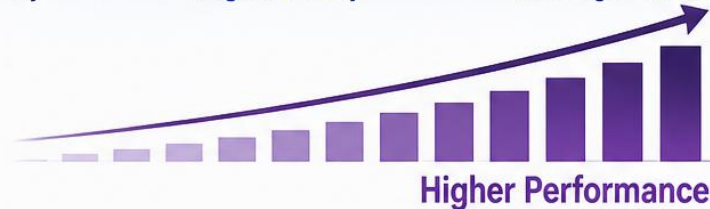
Smaller Size | More Devices | Higher Performance



Lower Density

Higher Density

Much Higher Density



Microelectronics trends

MICROELECTRONICS TRENDS



Higher degree of integration



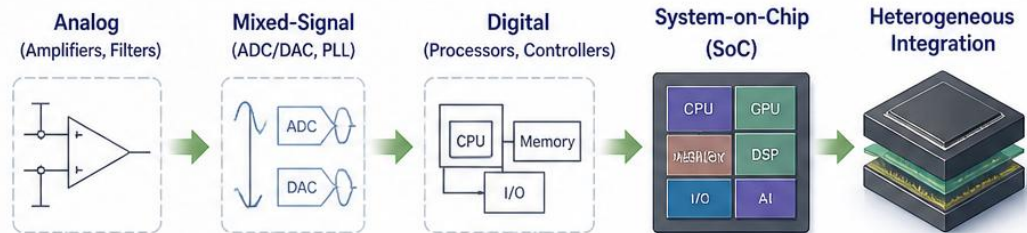
More Integration | More Functions | Smaller Size | Lower Cost



increased device reliability



inclusion of complex designs

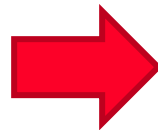
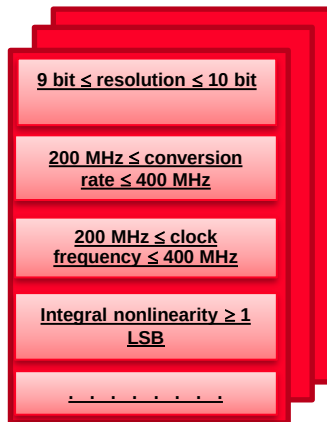


From Simple to Complex | More Capabilities | Greater Innovation

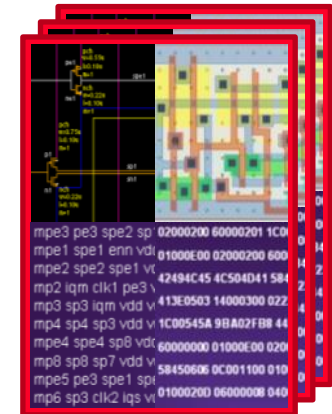
Electronic Design Process

- Design process is the process of getting IC design from specification.

Specification



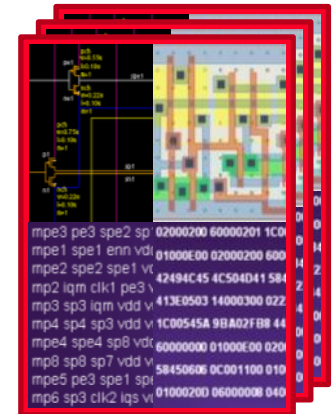
IC design



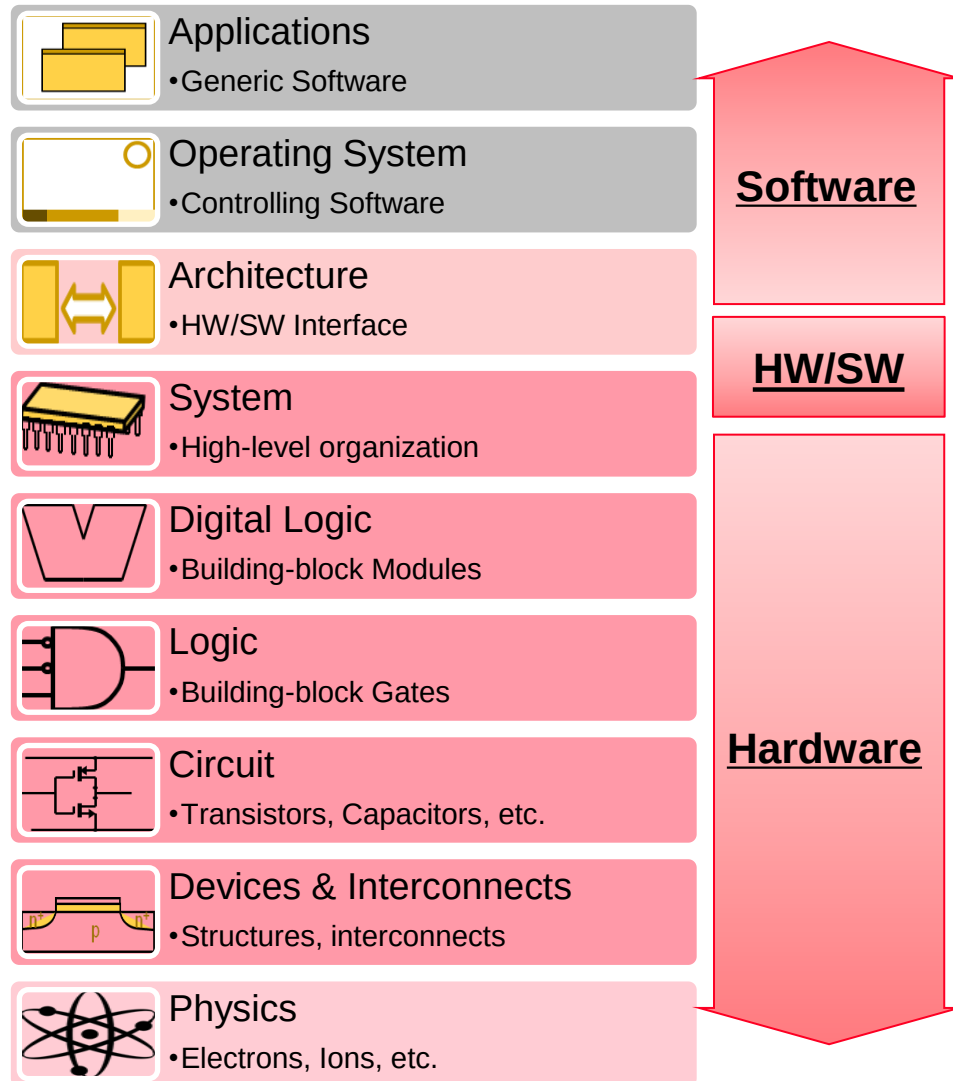
Electronic Design Process

- IC Characteristics
 - Area
 - Speed
 - Power dissipation
 - Design time
 - Testability
- It is impossible to design an integrated circuit at once while having required characteristics
- The complexity is simply too high

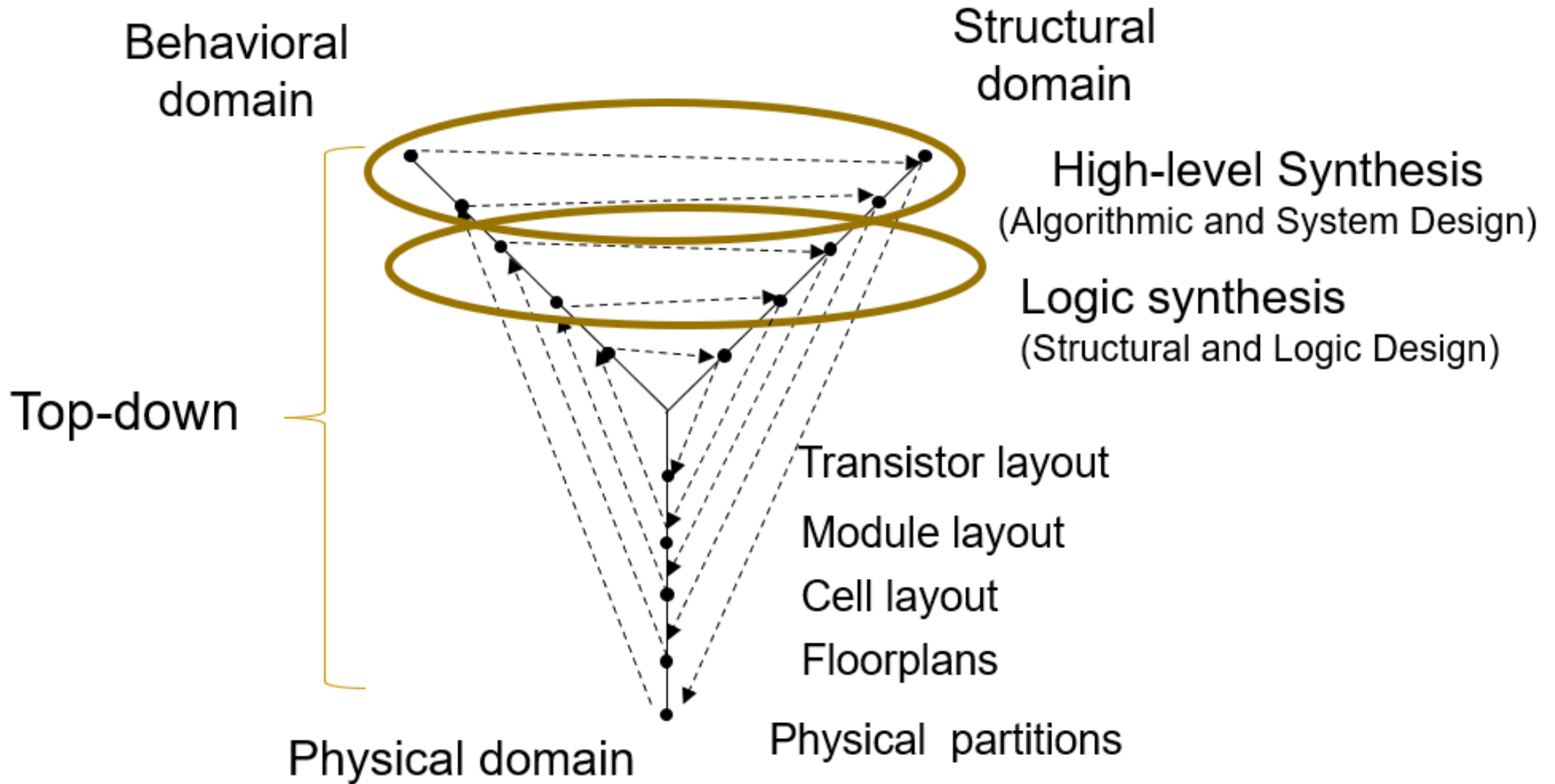
IC design



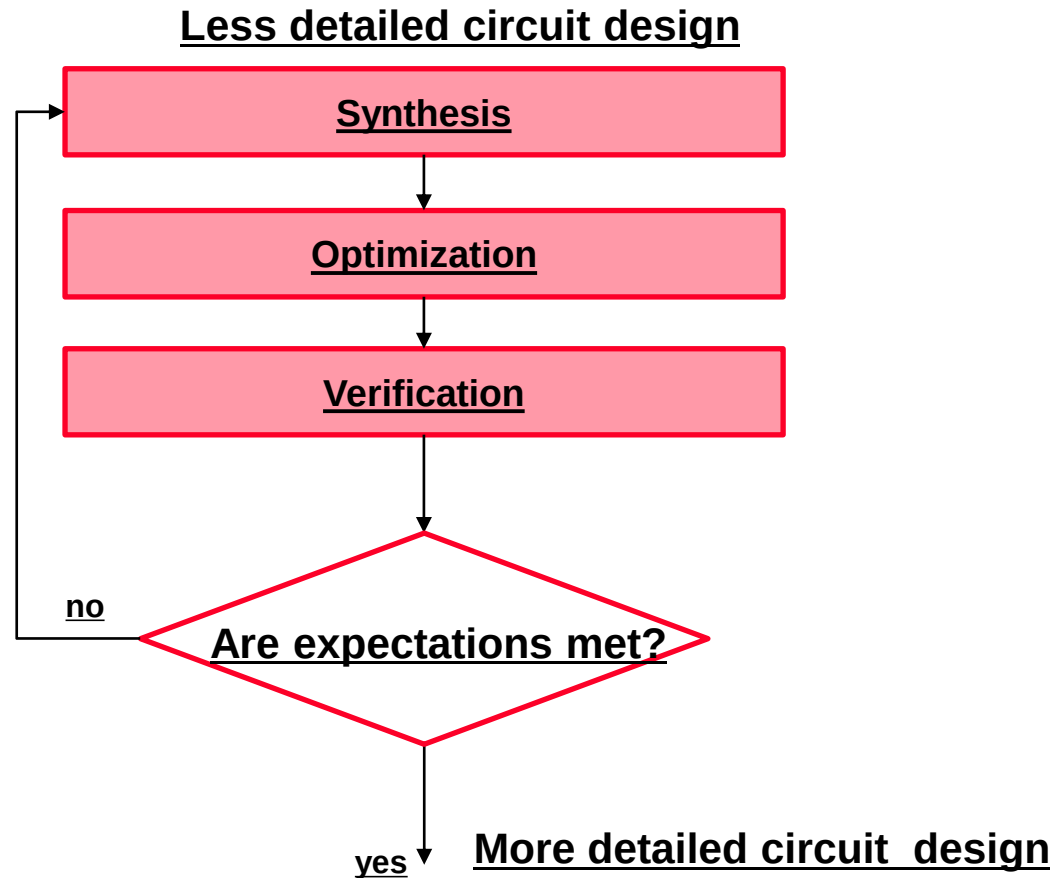
Design Abstraction Levels



Only Top-down Design Methodology



Problems Solved in One Design Level



Reference Slide

Thanks to *Synopsys University Courseware*

Digital Design Flow by Synopsys Tools



Synopsys University Courseware
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IC Design Flow
Lecture - 3
Developed By: Vazgen Melikyan

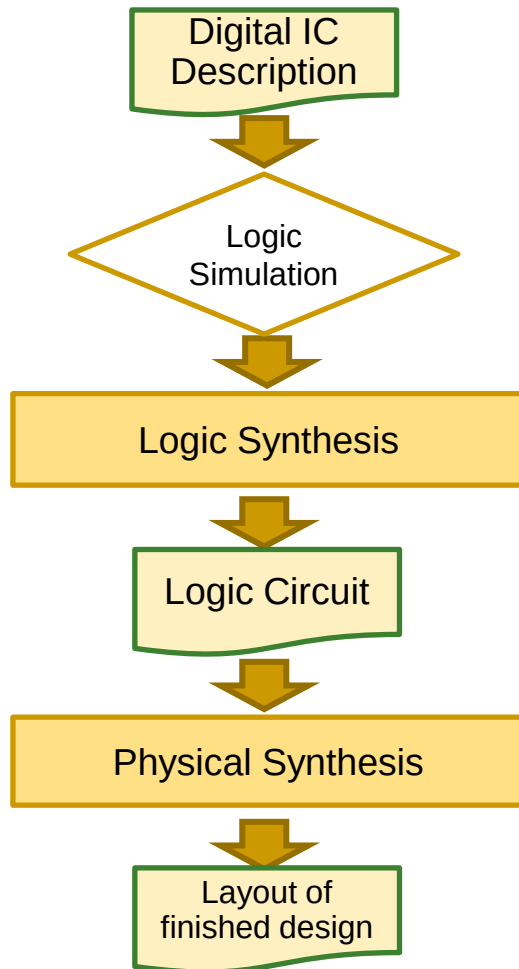


Basic Steps of Digital Design Flow

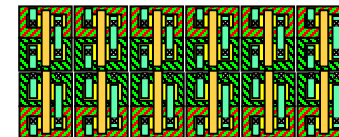
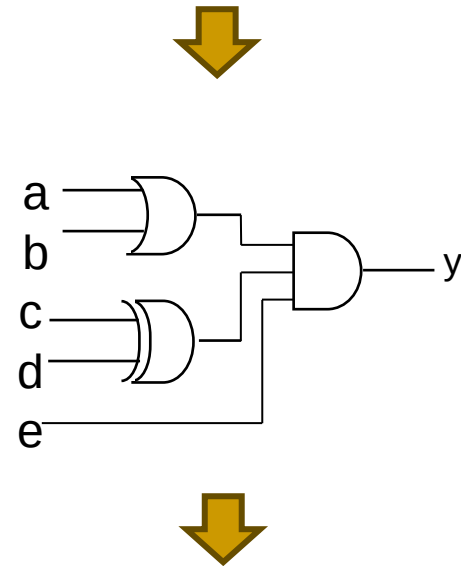
- Circuit design
 - Based on specification, synthesize the netlist
- Physical design
 - Based on the netlist, synthesize layout
- Verification
 - Perform pre-layout and post-layout verification of IC performance



Basic Steps of Digital Design



$$y = (a + b) \& (c \oplus d) \& e$$



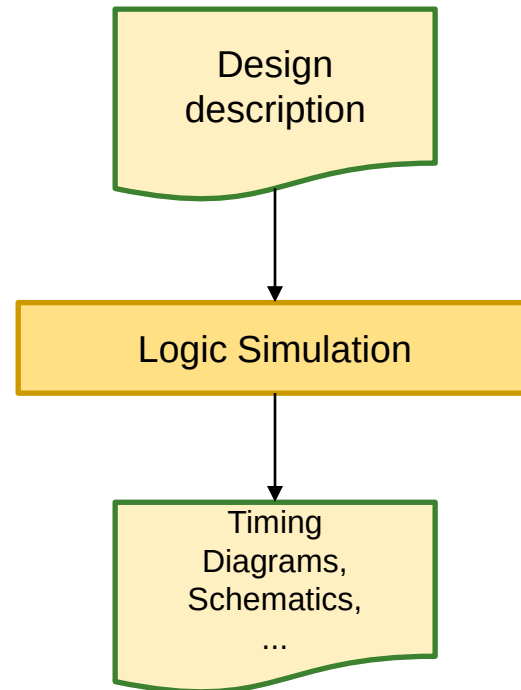
Digital IC Description

- Description of Digital IC functionality
- With the help of Verilog or VHDL in the specification
- An example of specification line:
 - if incoming_call AND line_is_available then RING



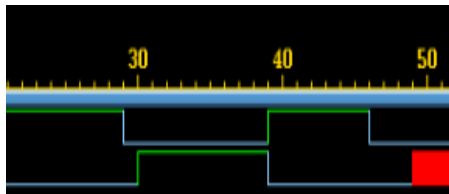
Logic Simulation

- Verifying functionality and timing through Simulation and Testbench Generation (example: VCS-MX, VERA)



Logic Simulation Steps

Compiling is the first step to simulate your design. In this phase the instance hierarchy is built and a binary executable is generated.



Design Verilog description



Compilation



binary executable file



Simulation



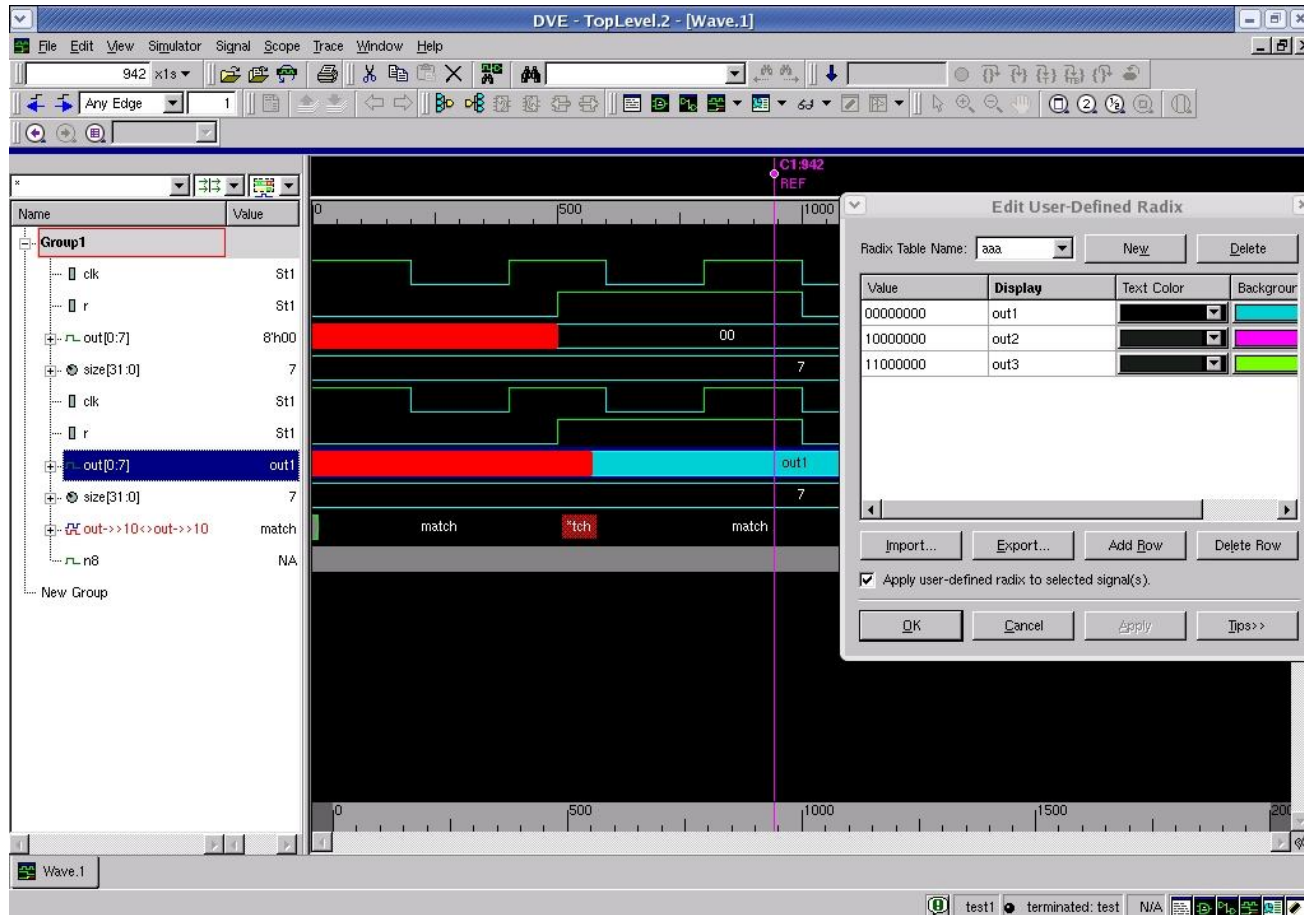
Diagrams

```
Module test(out,in,clk);  
  input in,clk;  
  output out;  
  wire a;  
  dff dff0(a,in,clk);  
  ...
```

Simulation phase uses the binary executable file to simulate your design.

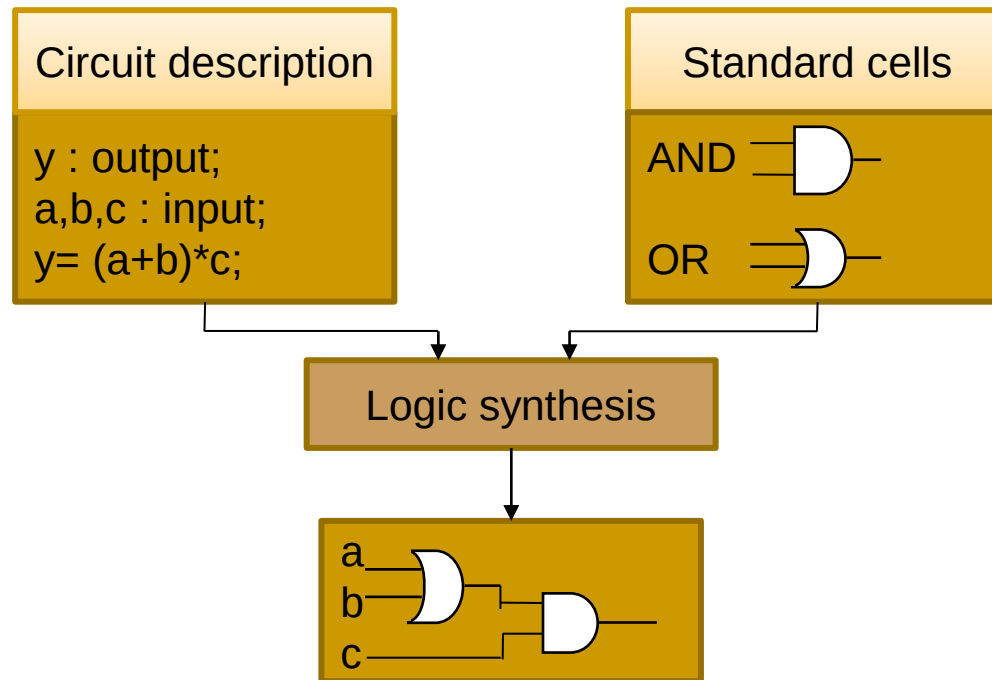


VCS



Logic Synthesis

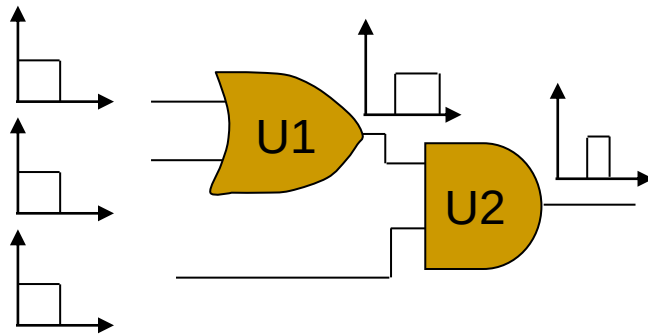
- Logic synthesis is the process which produces logic circuit from circuit description



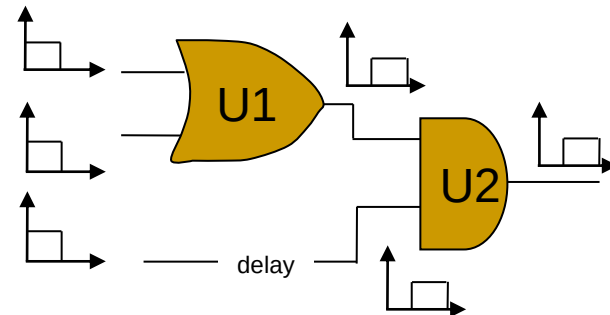
Logic Optimization

- Logic synthesis also optimizes the circuit. The problem:
 - Circuit simply created from function can possibly operate not as expected.

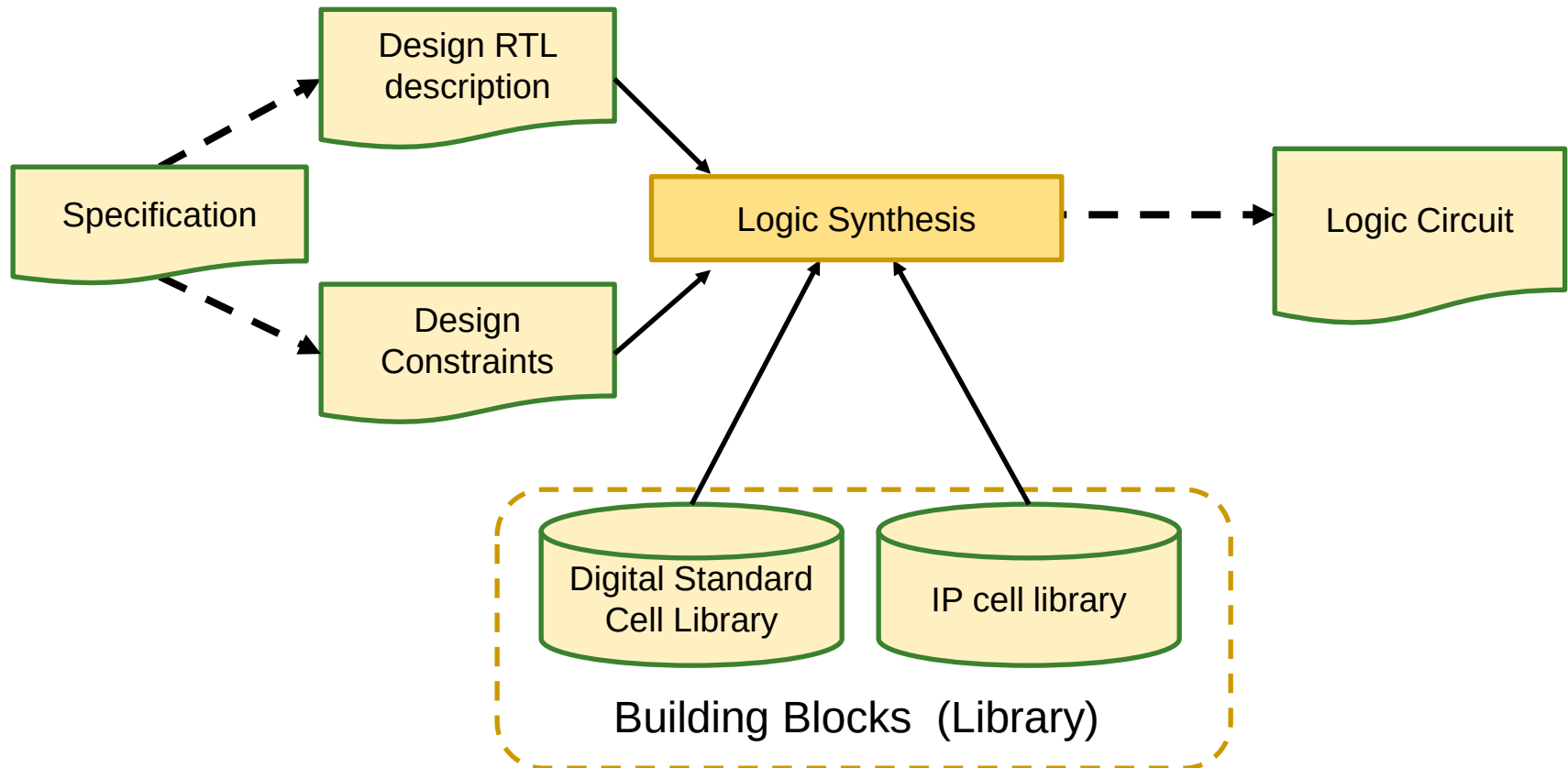
The delay of U1 element will affect final result



Additional elements should be added to the circuit to ensure correct operation

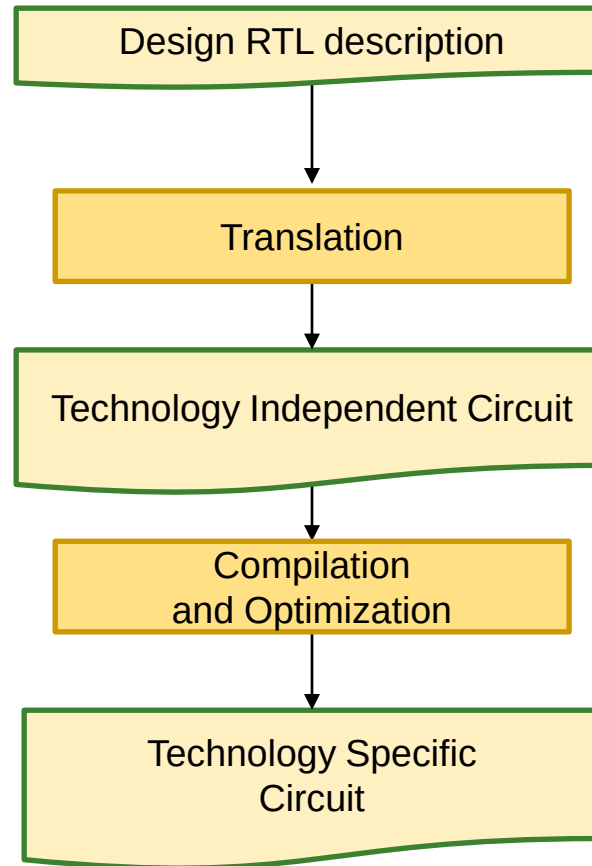
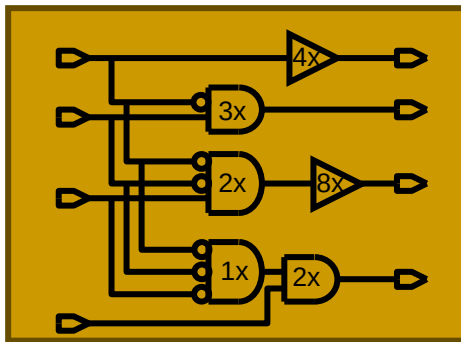


Logic Synthesis Data Flow

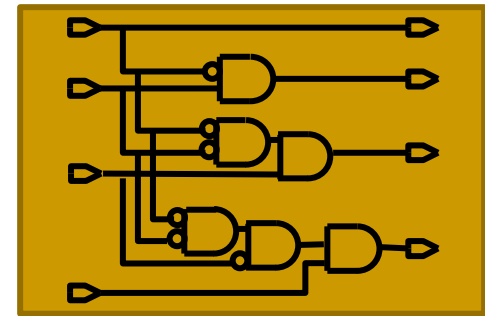


Logic Synthesis Steps

Technology Specific Circuit is get from independent one by replacing all components by real blocks (standard cells). This replacement process is also called mapping



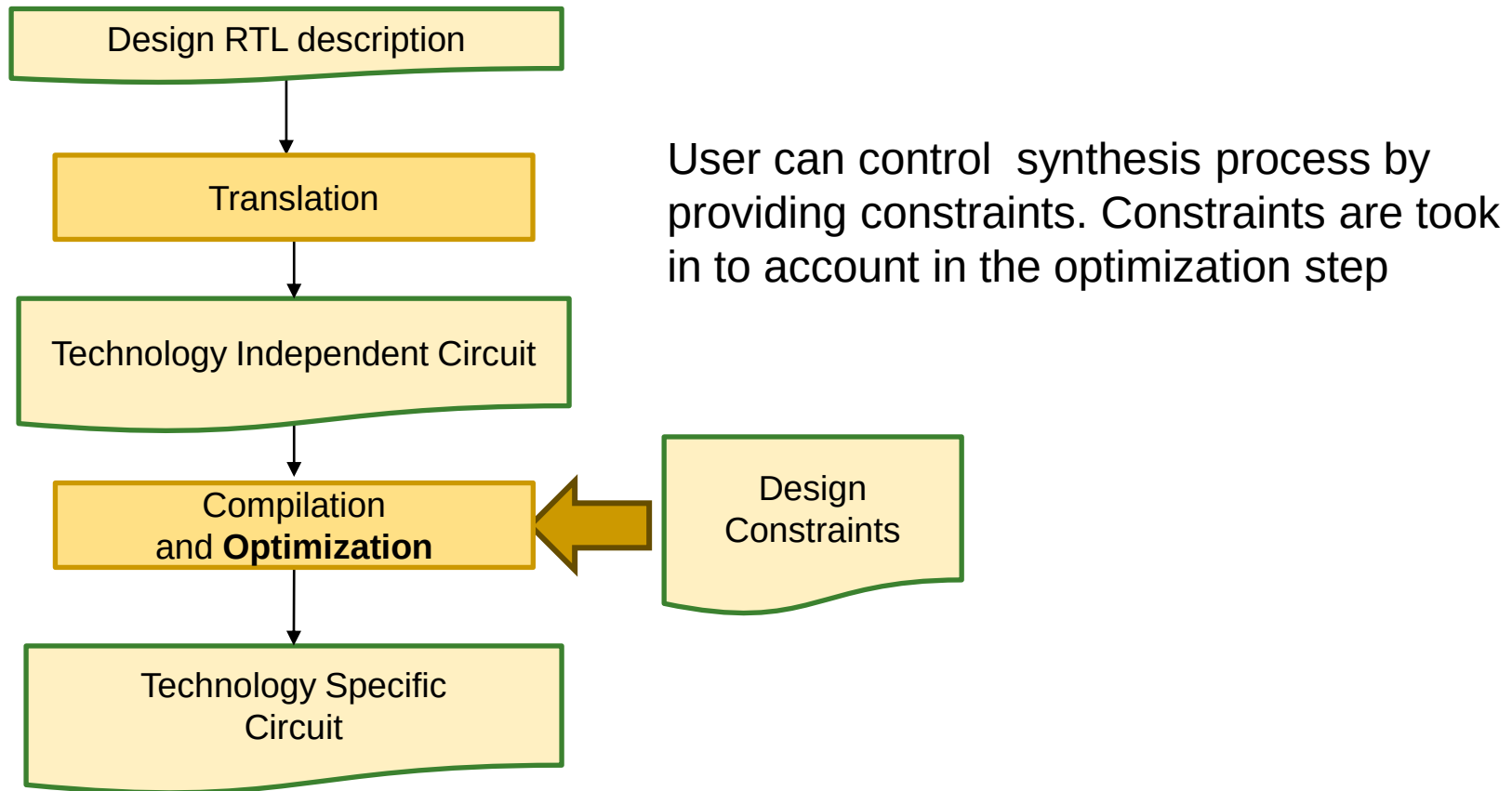
```
residue = 16'h0000;  
if (high_bits == 2'b10)  
    residue = state_table[index];  
else  
    state_table[index] = 16'h0000;
```



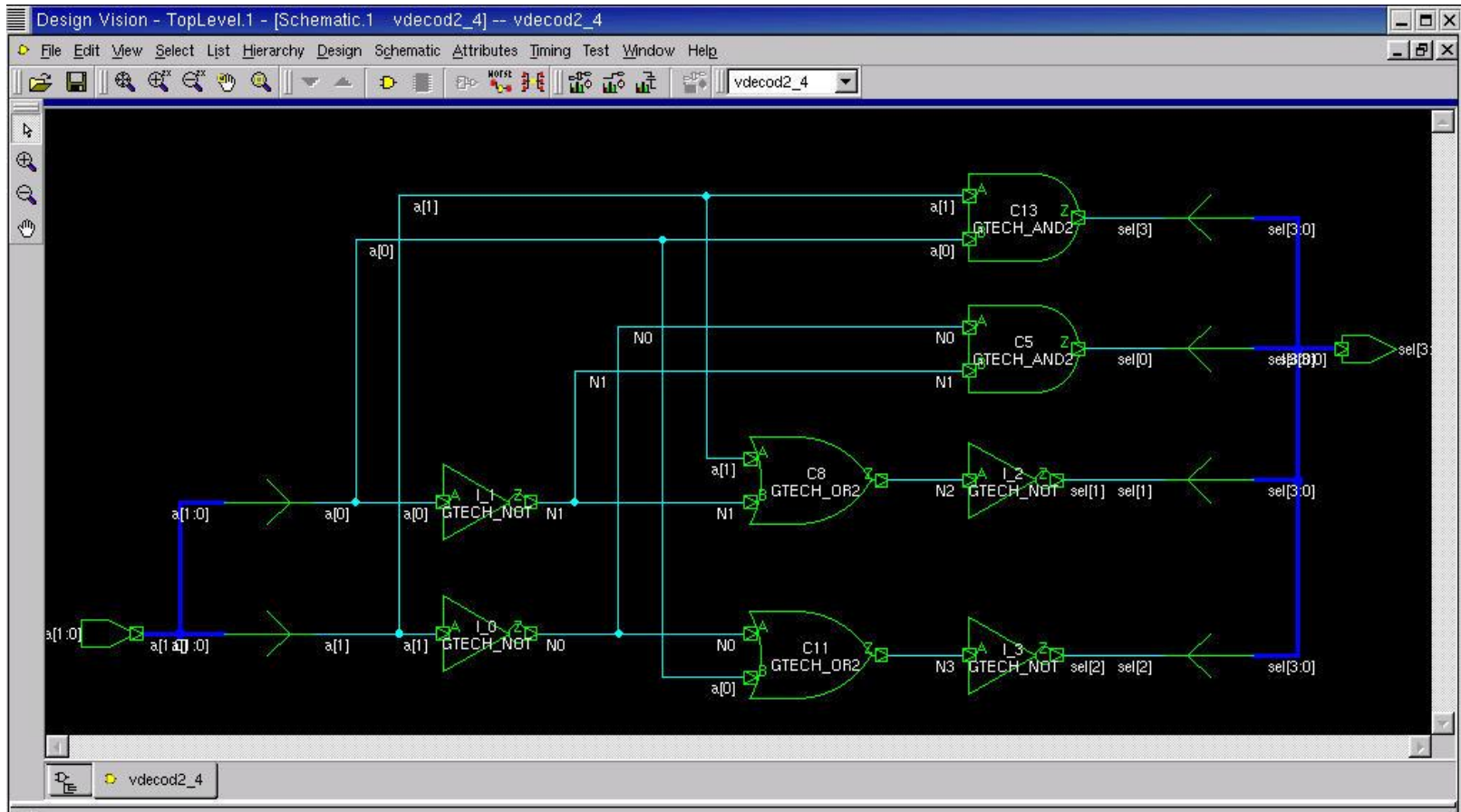
Technology Independent Circuit is logic circuit which fully implements function described but is built from **Generic Boolean Gates**.



Constraint-Driven Synthesis

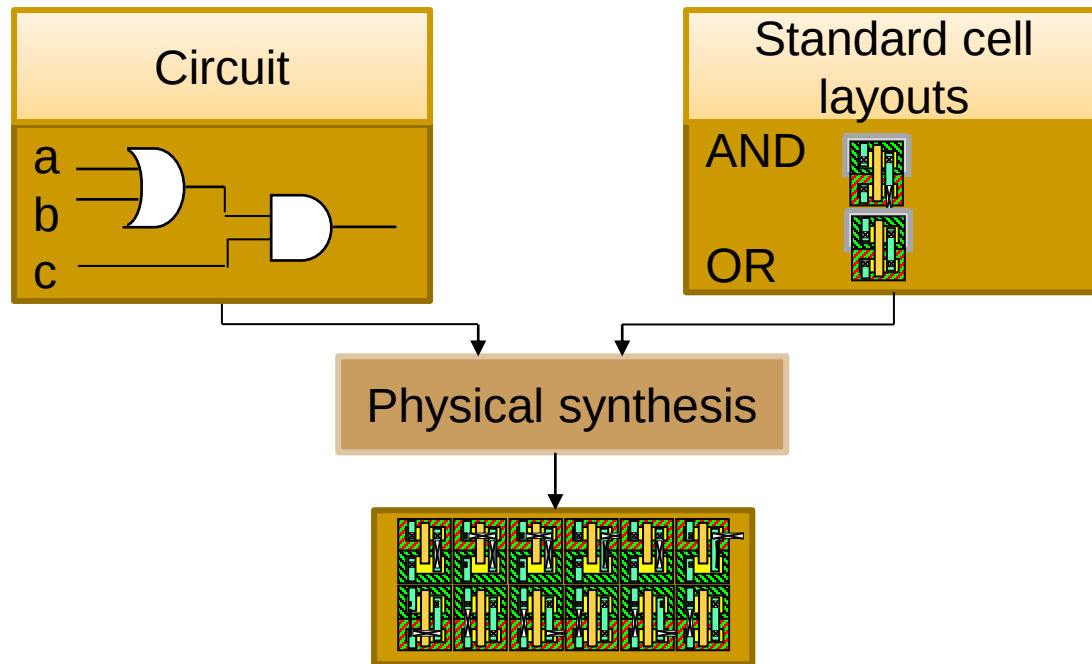


Design Compiler

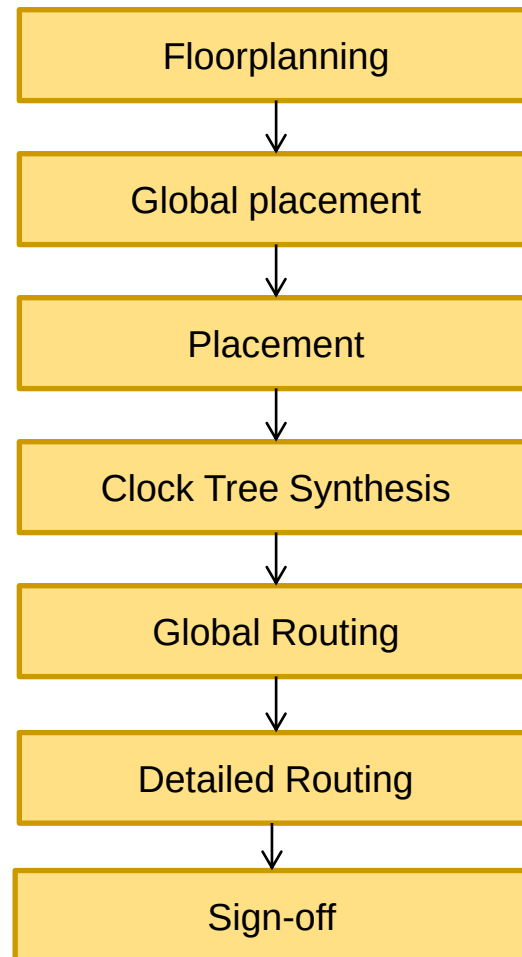


Physical Synthesis

- Physical synthesis is the process that produces layout of logic circuit.

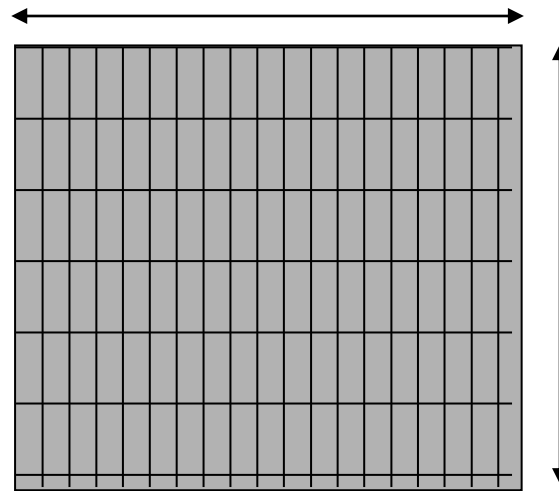


Physical Synthesis Steps



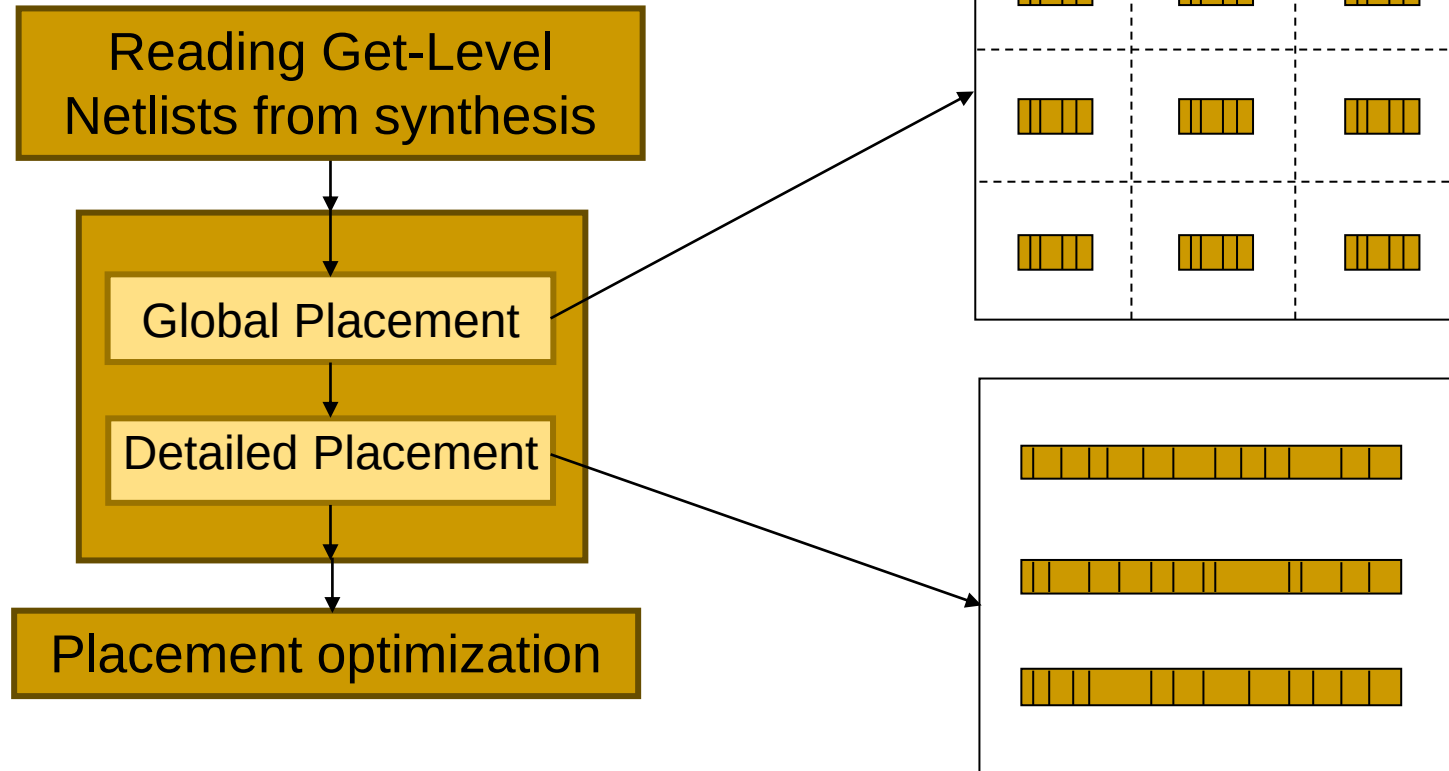
Floorplanning

- During the floorplanning step the overall cell is defined, including: cell size, supply network, etc.



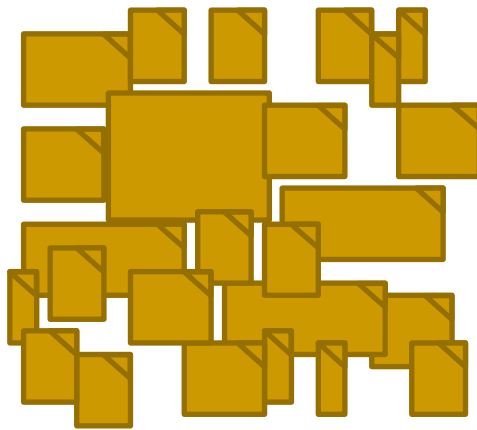
Floorplan

Global Placement (2)

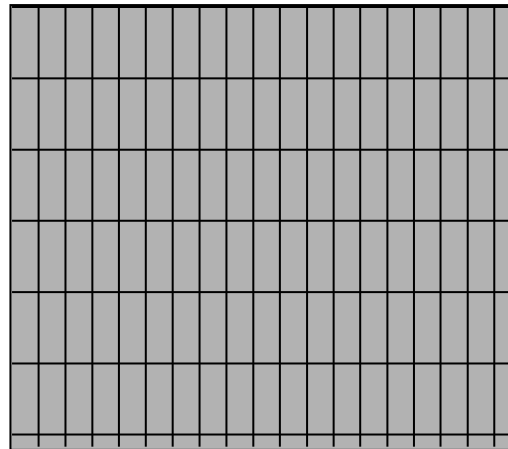


Placement

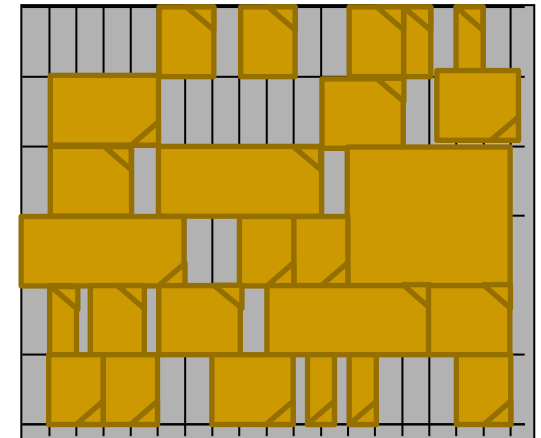
- Placement – exact placement of modules (modules can be standard cells, IPs).
- The goal is to minimize the total area and interconnect length



Cells from a
circuit

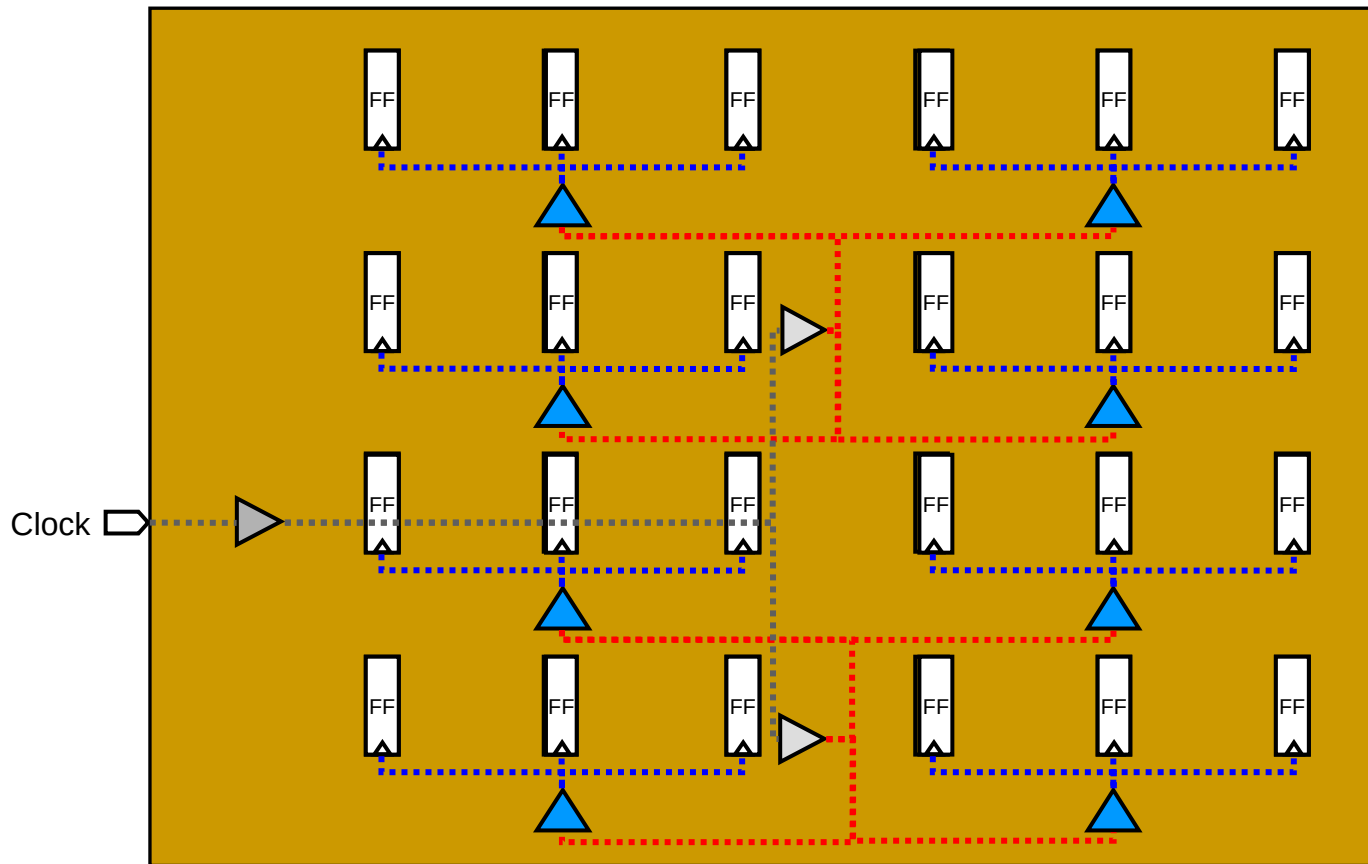


Floorplan



Placed
design

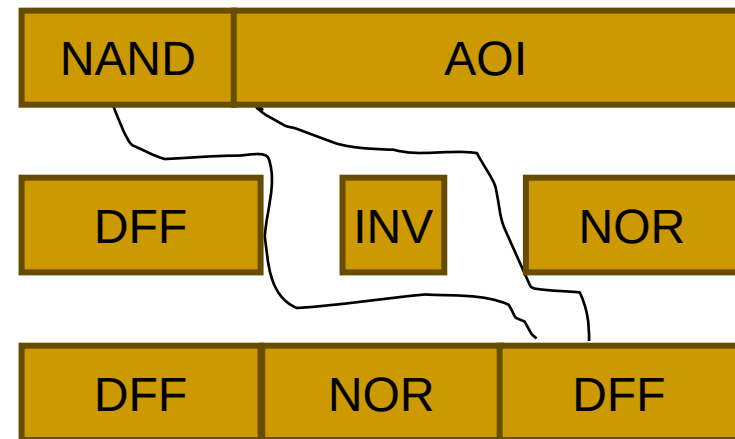
Clock Tree Synthesis (CTS)



A buffer tree is built to balance the loads and minimize the skew

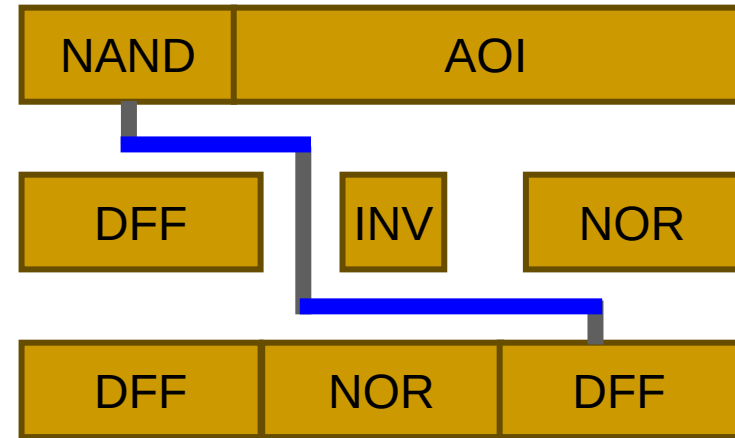
Global Routing

- Determining overall path of all routes
- Picking channels to route through
- Seeking to reduce delay, channel widths

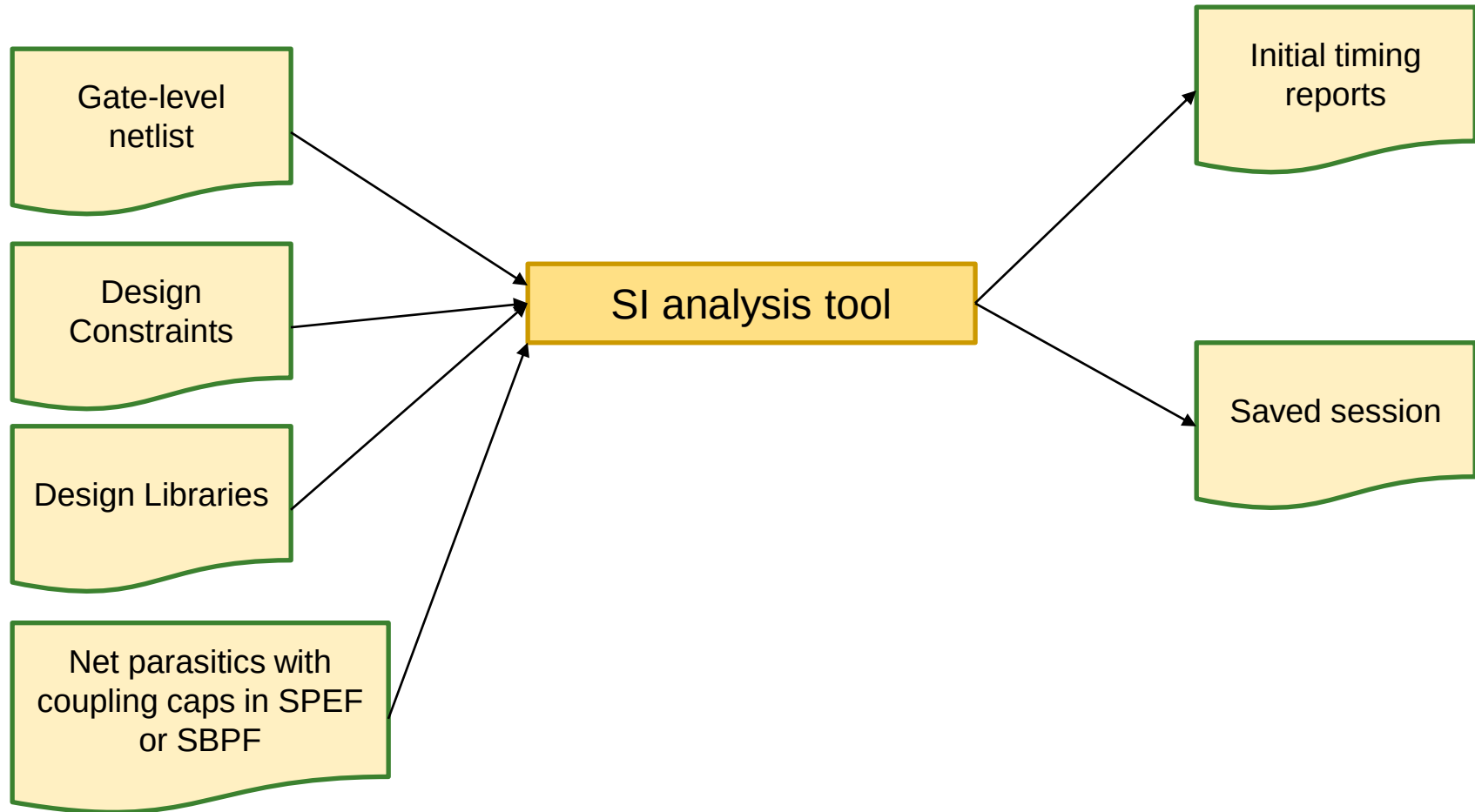


Detailed Routing

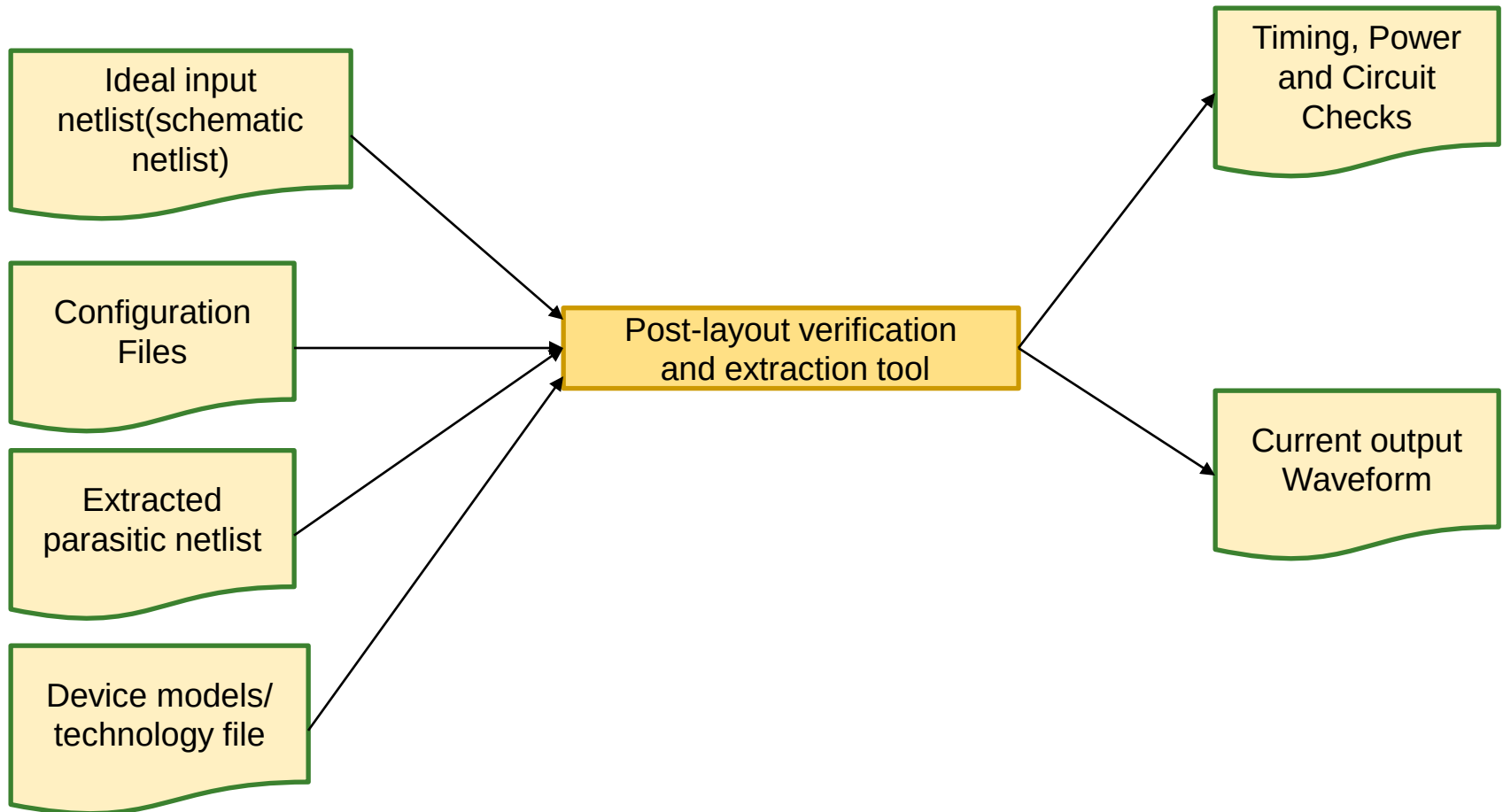
- Determining exactly how each signal is routed through each region
- Seeking to reduce routing area



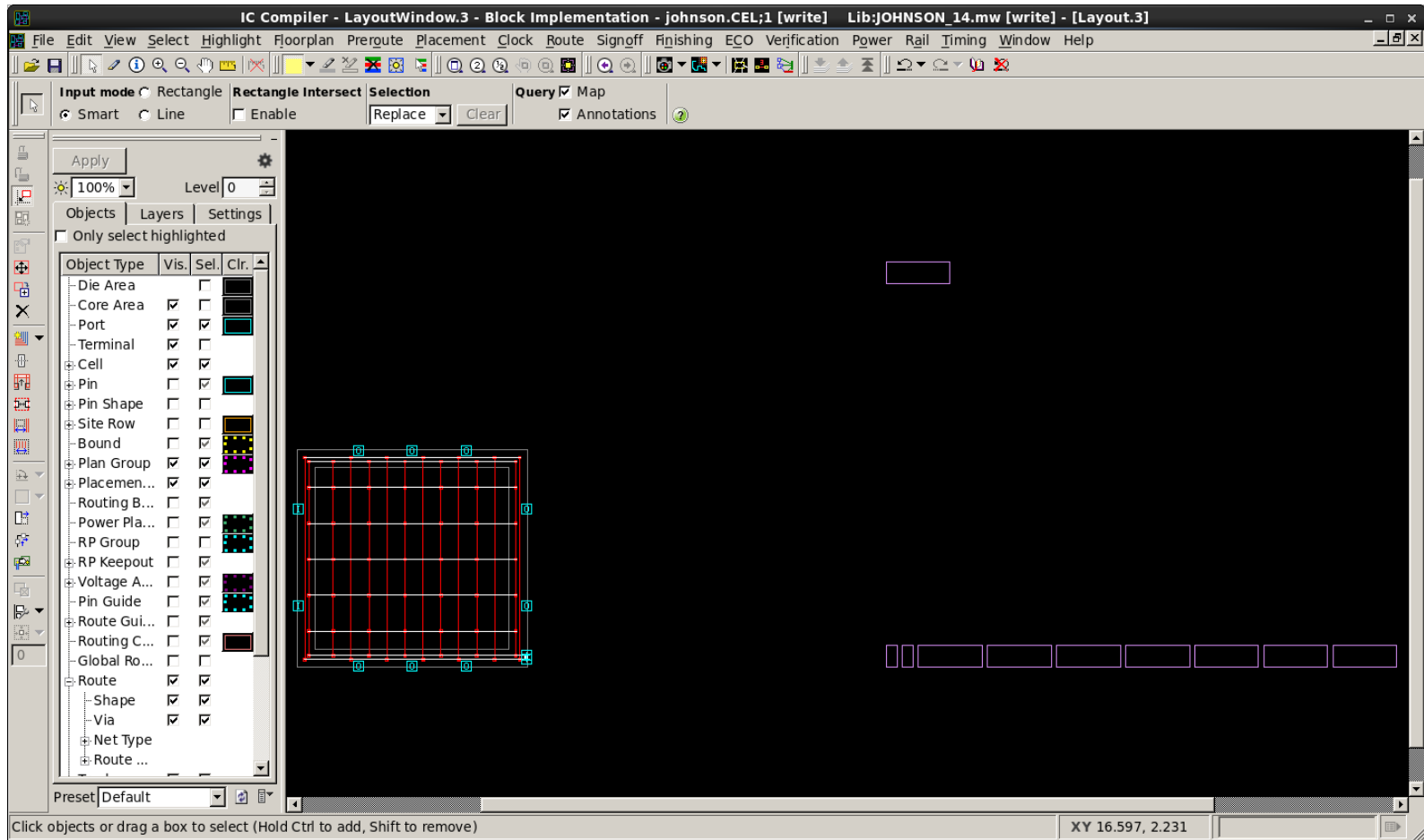
Sign-off: SI Analysis



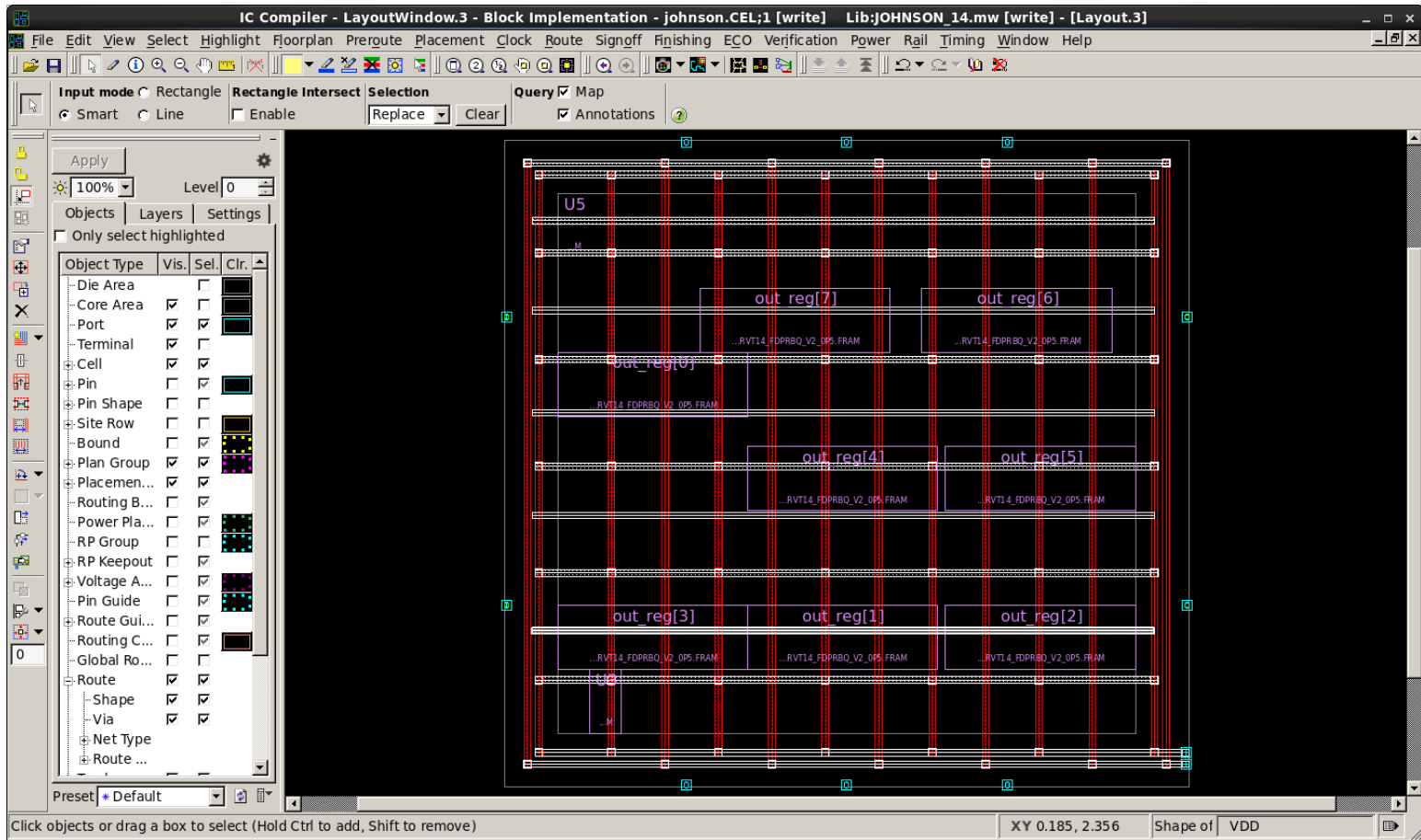
Sign-off: Extraction and Verification



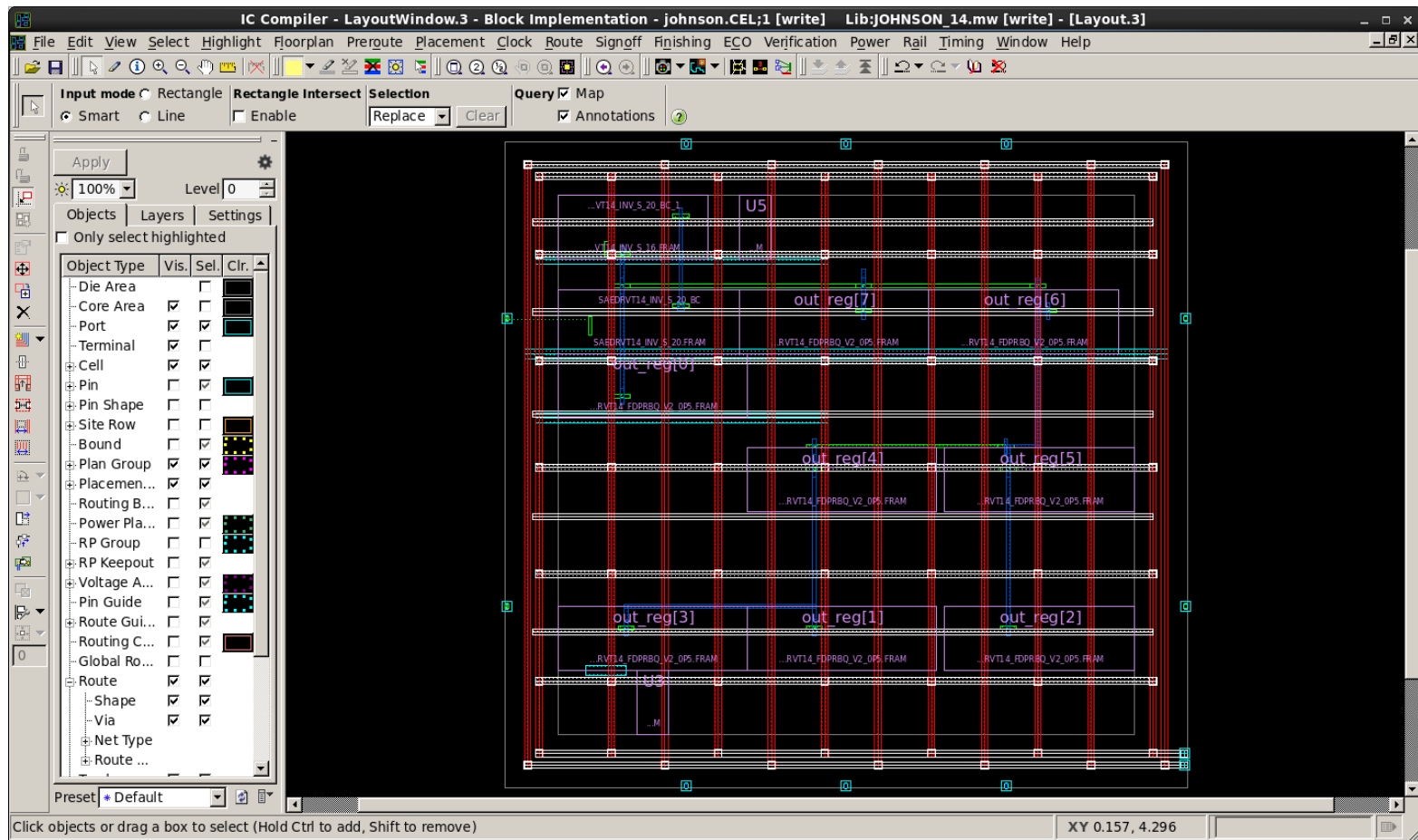
IC Compiler: Floorplanning



IC Compiler: Placement



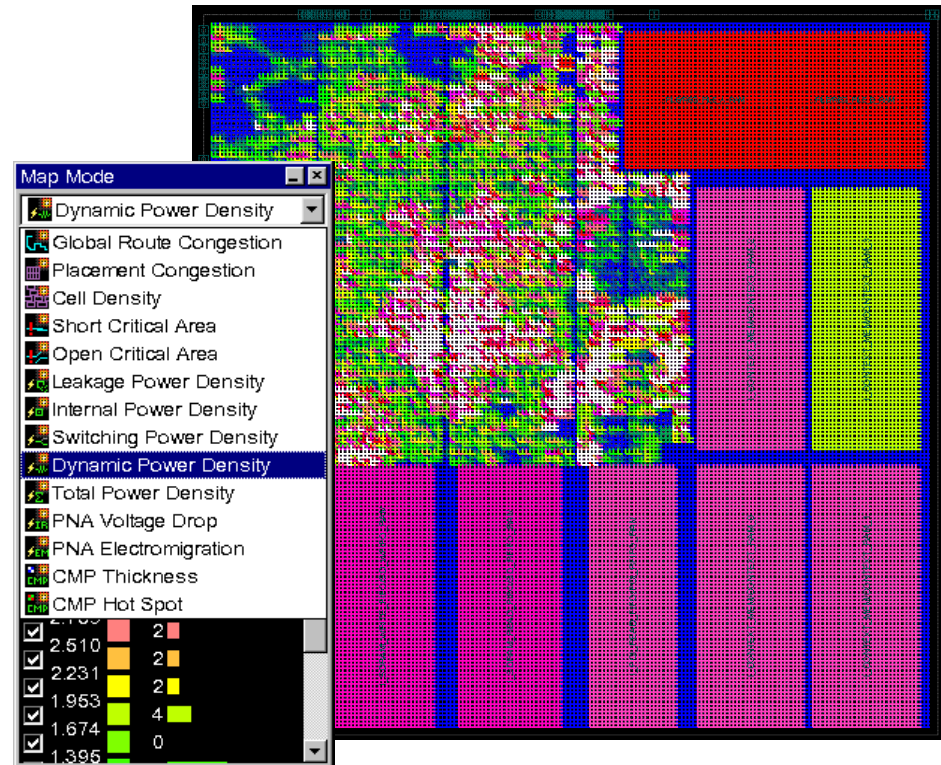
IC Compiler: CTS



SYNOPSYS®

IC Compiler: Power Density Map

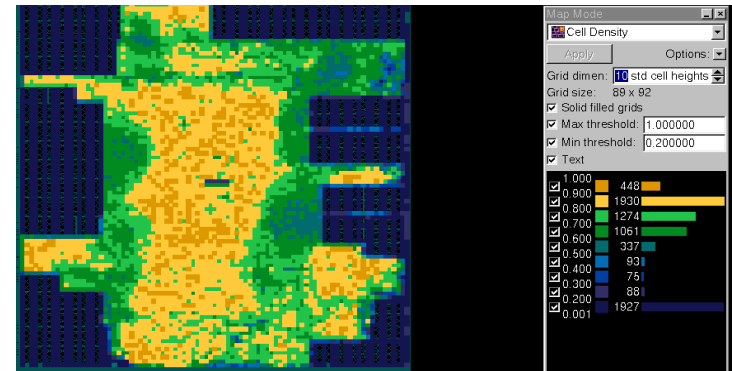
- Ideally, power dissipation should be distributed evenly across the core area
- High power density in a small area can lead to
 - ❑ IR-drop
 - ❑ EM
 - ❑ Poor product life



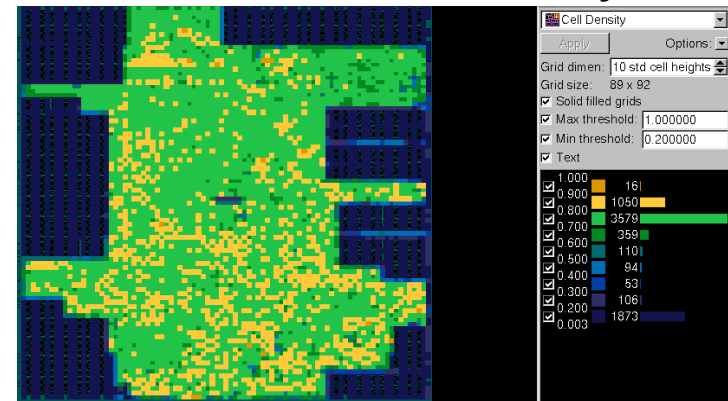
IC Compiler: Congestion Map

- No need to use -congestion unnecessarily
 - By default, physical synthesis tool performs some congestion optimization that has a reasonable chance of providing acceptable congestion
 - Congestion increases efforts of algorithm of a congestion
 - On average –congestion increases runtime by 20%
 - For better correlation to post-route, congestion-driven placement is enabled based on GR congestion map

Causes high local utilization



Gives uniform density





New Challenges to IC Design Flow

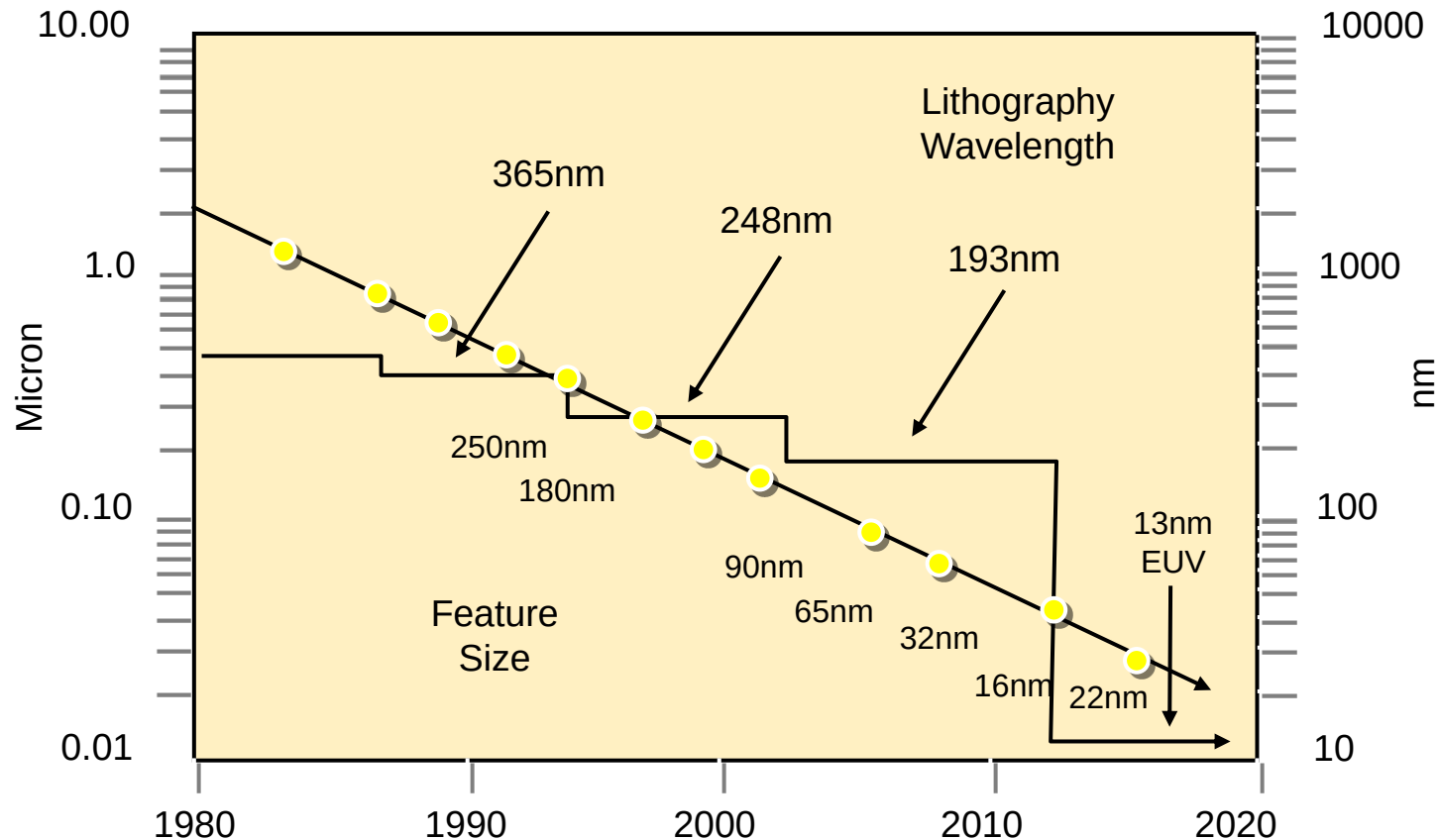


Implementation of Chips

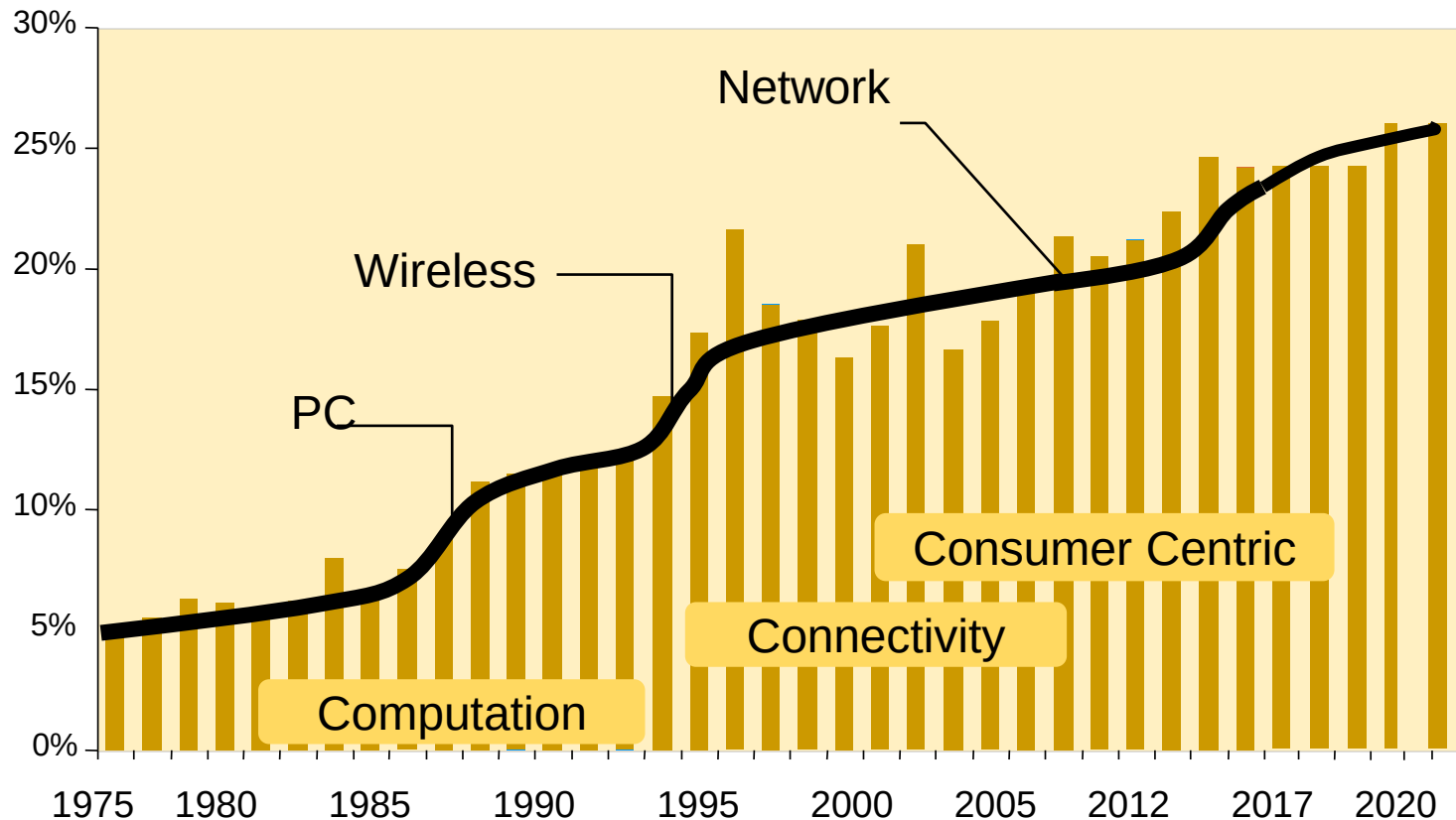
- Semiconductor industry is about \$250B.
- Electronic Design Automation (EDA), i.e. the industry of creating automated design tools of semiconductor chips is about \$4B.
- Semiconductor chips are used in:
 - ❑ Computers
 - ❑ Cellular phones
 - ❑ Gaming systems
 - ❑ DVD players, TVs
 - ❑ Watches
 - ❑ Cars
 - ❑ Medical devices
 - ❑ Pacemakers and coffee pots
 - ❑ Space stations
 - ❑ Greeting cards
 - ❑ . . .



Lithography Trends



The Cost of Chip Content in Electronic Systems



History and Evolution of the IC Industry

Date	1999	2007	2015	2017	2020
Technology (nm)	180	65	14	7	5
Minimum mask count	22/24	25	27/28	29/30	35/36
Wafer diameter (mm)	200	400	450	450	300
Memory samples (bits)	1G	8G	1T	1T	1T
Transistors /cm ²	6.2Ø	180M	300M	390M	489M
Maximum number of metal layers	6-7	9	13	13	15
Clock frequency (MHz)	1250	3200	10000	10000	10000
IC sizes (mm ²)	400	760	1940	2240	2970
Power supply (V)	1.5-1.6	0.8-1.2	0.6-0.8	0.37-0.42	0.25-0.32
Maximum power (W)	90	150	167	183	215
Number of pins	700	1957	2900	3350	3850



History and Evolution of the IC Industry (2)

Other source

Date	2004	2007	2010	2013	2017	2020	2025
Technology (nm)	90	65	45	32	7	5	3
Clock frequency (MHz)	4171	9285	15079	22980	39683	53207	89345
Chip area (mm ²)	280	280	280	280	280	280	290
Supply voltage (V)	1.2	1.1	1.0	0.9	0.8	0.7	0.4
Maximum power (W)	158	189	218	251	288	300	350
Average chip current (A)	132	172	218	279	360	429	480



Technology Roadmap

.7μm	.5μm	.35μm	.2μm	.25μm	.18μm	.13μm	90nm	65nm	45nm	32nm	14nm	7nm	5nm
1992	1995	1997	1999	2000	2002	2003	2004	2007	2010	2013	2015	2017	2020

Smaller, Cheaper

Increasing Design Cost

Faster

Parasitics
CD Variability

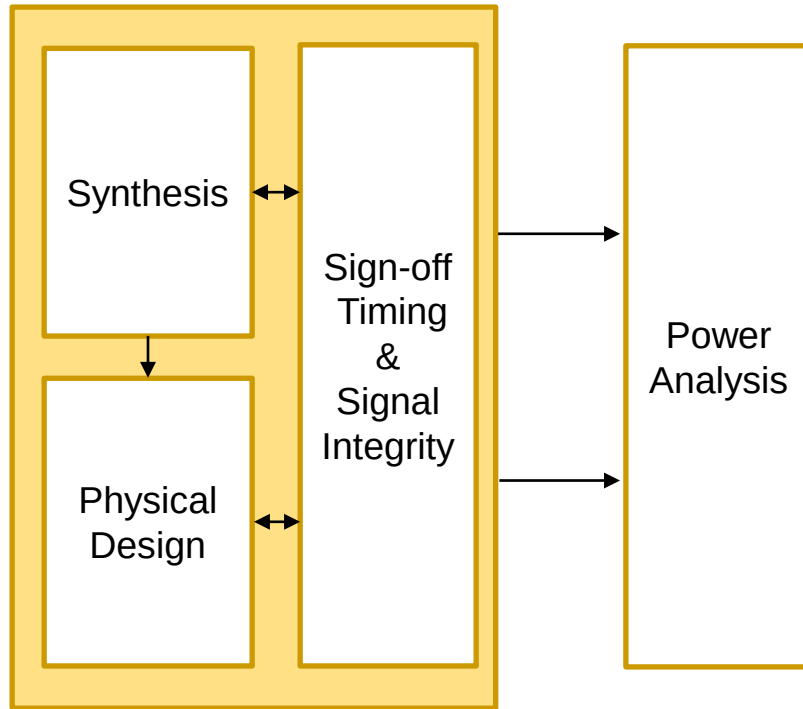
Less Power

Leakage
Power Density

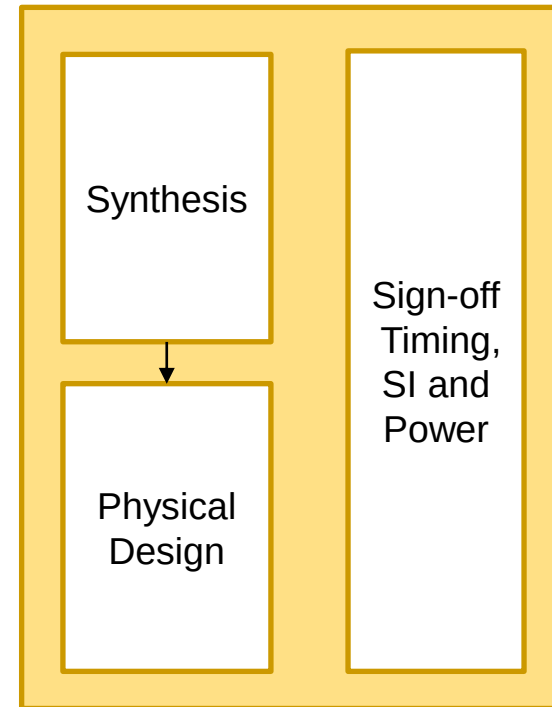


Concurrent Design Methodology

Traditional



Concurrent

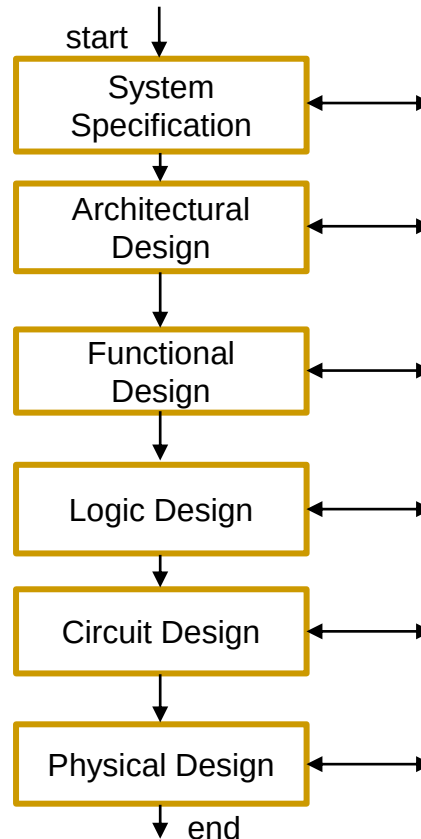


New Challenges in the Design Flow

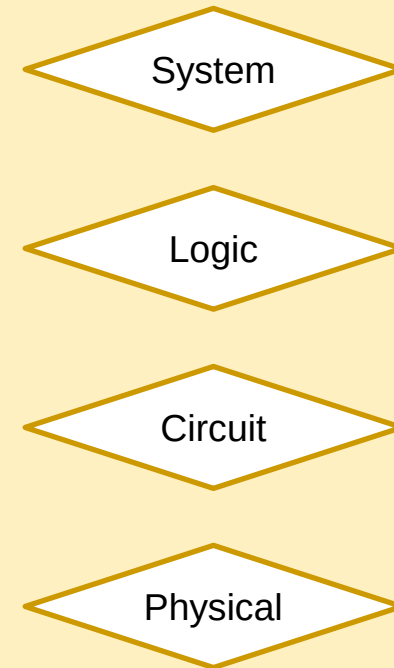
■ Design Challenges

- IP
- Timing closure
- Signal integrity
- Leakage power
- Dynamic power
- Test
- Yield
- Printability

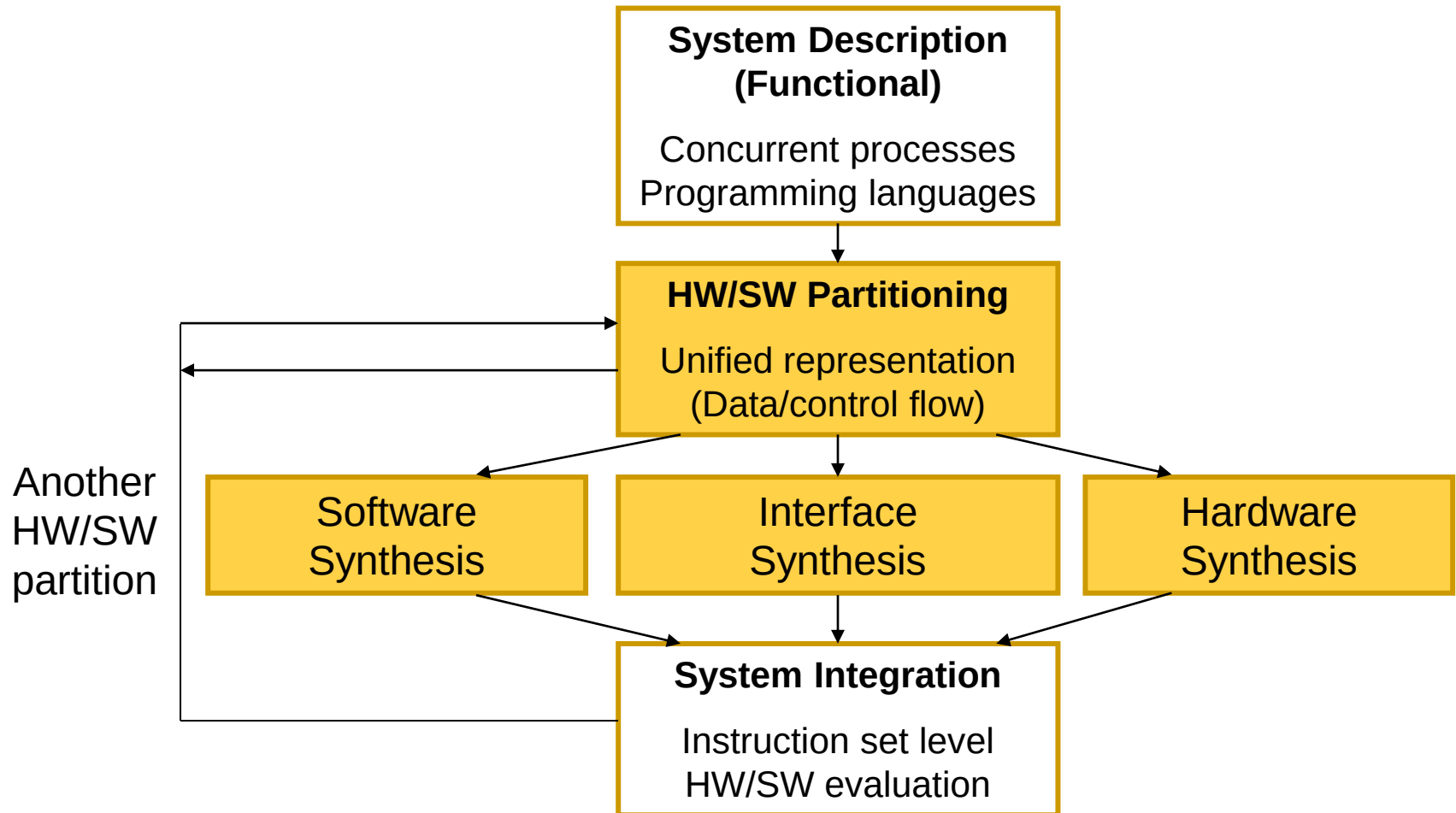
Design Implementation



Design Verification



HW/SW Co-design Flow





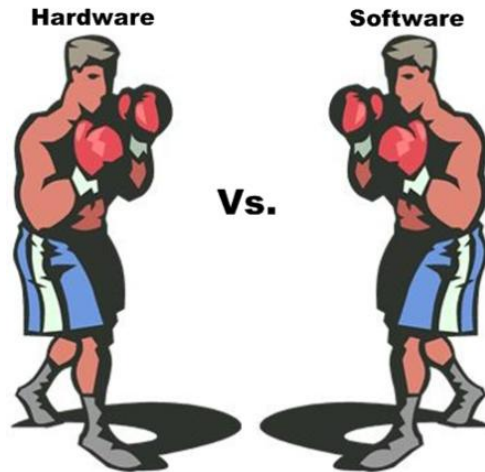
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Digital Systems

Judged by its objectives in application domain

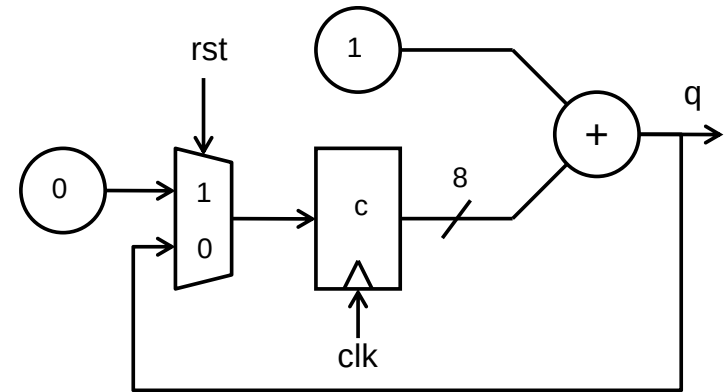
- Performance (Speed, Power)
- Design and Manufacturing cost (Cost, time)
- Ease of Programmability

It depends on both the hardware and software components

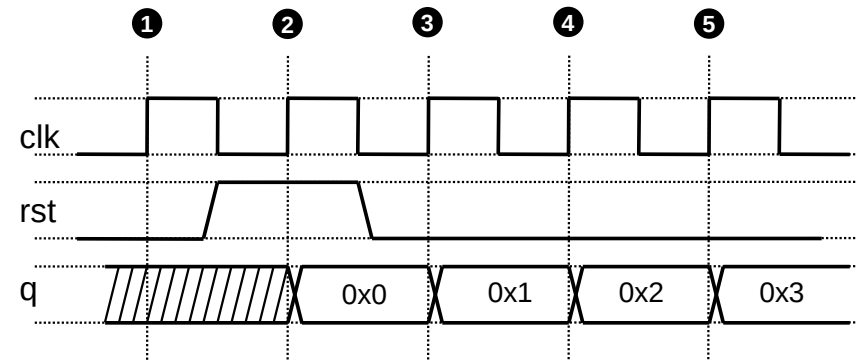


Hardware

- Single-clock synchronous digital circuits
- Cycle-based hardware modeling (RTL)
 - Registers,
 - Adders
 - Multiplexers



```
dp count(in rst: ns(1)) {  
    reg c : ns(8);  
    always {  
        q = c;  
        c = rst ? 0 : c + 1;  
    }  
}
```



Software

- Organization of variables into memory
- Techniques to control this from within a high-level programming language such as C
- Single-thread sequential programs

Listing 1.1 C example

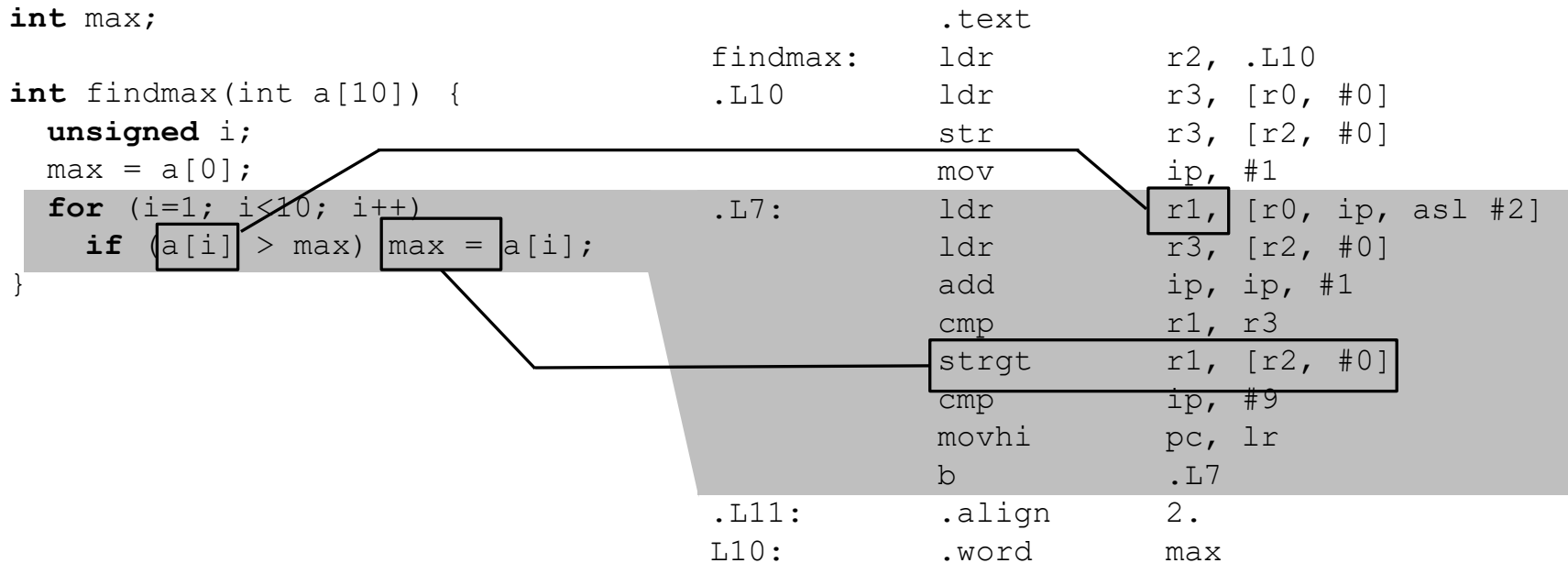
```
1  int max;
2
3  int findmax(int a[10]) {
4      unsigned i;
5      max = a[0];
6      for (i=1; i<10; i++)
7          if (a[i] > max) max = a[i];
8  }
```

Software

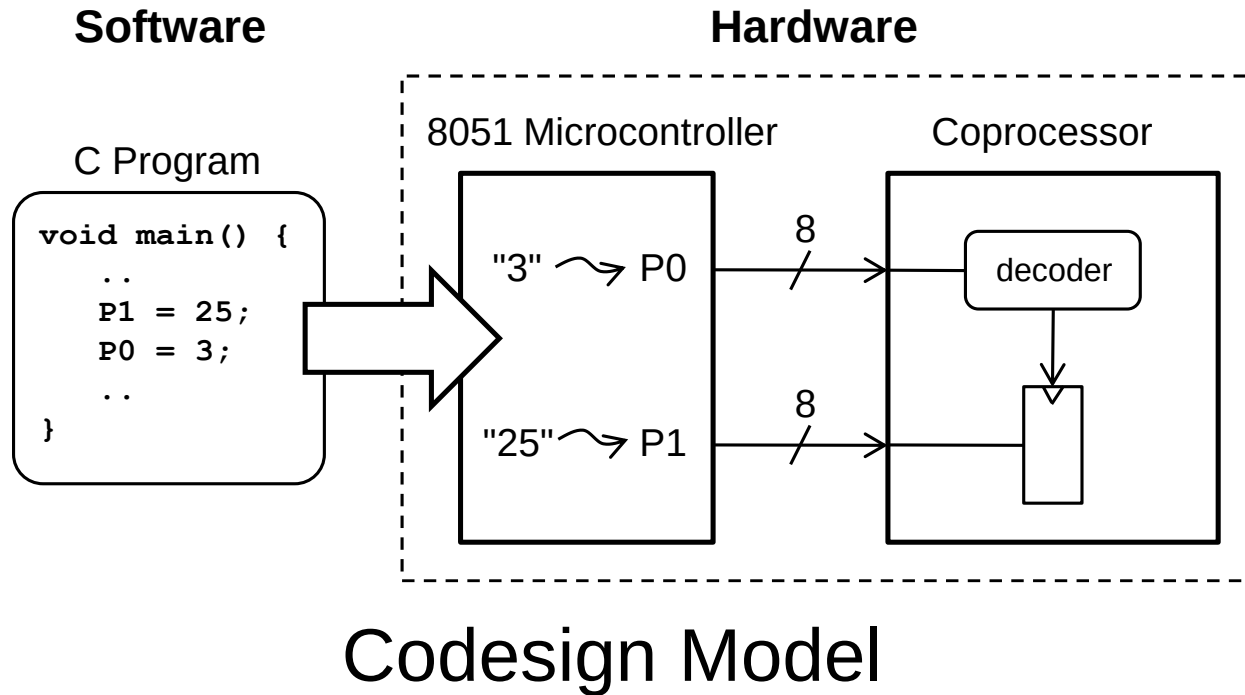
```
int max;

int findmax(int a[10]) {
    unsigned i;
    max = a[0];
    for (i=1; i<10; i++)
        if (a[i] > max) max = a[i];
}

                                .text
findmax:                        ldr        r2, .L10
                                ldr        r3, [r0, #0]
                                str        r3, [r2, #0]
                                mov        ip, #1
                                .L7:      ldr        r1, [r0, ip, asl #2]
                                ldr        r3, [r2, #0]
                                add        ip, ip, #1
                                cmp        r1, r3
                                strgt      r1, [r2, #0]
                                cmp        ip, #9
                                movhi     pc, lr
                                b          .L7
                                .L11:     .align    2.
                                L10:      .word    max
```

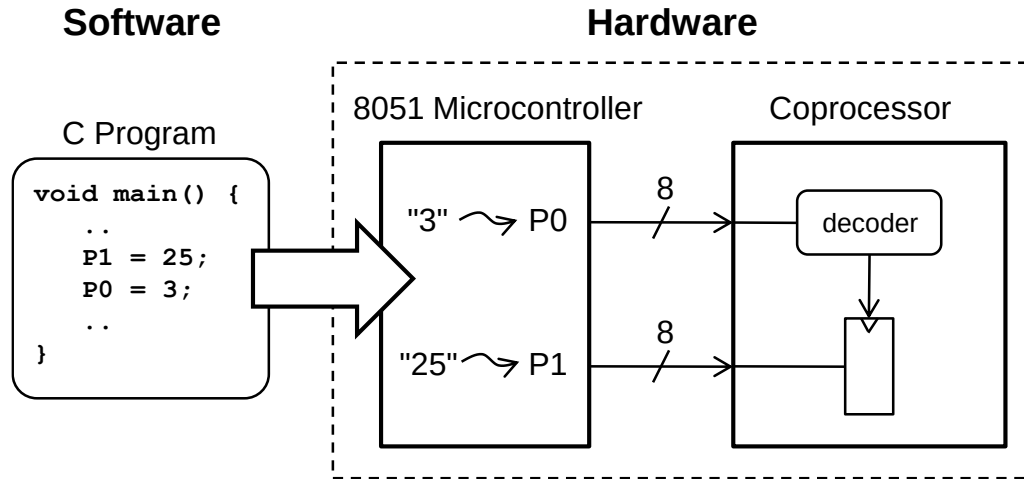


HW/SW Codesign



Meeting System level objectives by exploiting the synergism of hardware and software through their concurrent design

HW/SW Codesign



Hardware/Software Codesign is the design of cooperating hardware components and software components in a single design effort

HW/SW Codesign

- FPGA (Field Programmable gate Array)
 - Reconfigured to a user-specified netlist of digital gates
 - An FPGA's program is a “bitstream” => configure the netlist topology
 - **Writing “software” for an FPGA ?**
 - **Soft-core ?**



HW/SW Codesign

- DSP (Digital-Signal Processor)
 - Specialized instruction-set
 - Optimized for signal-processing applications

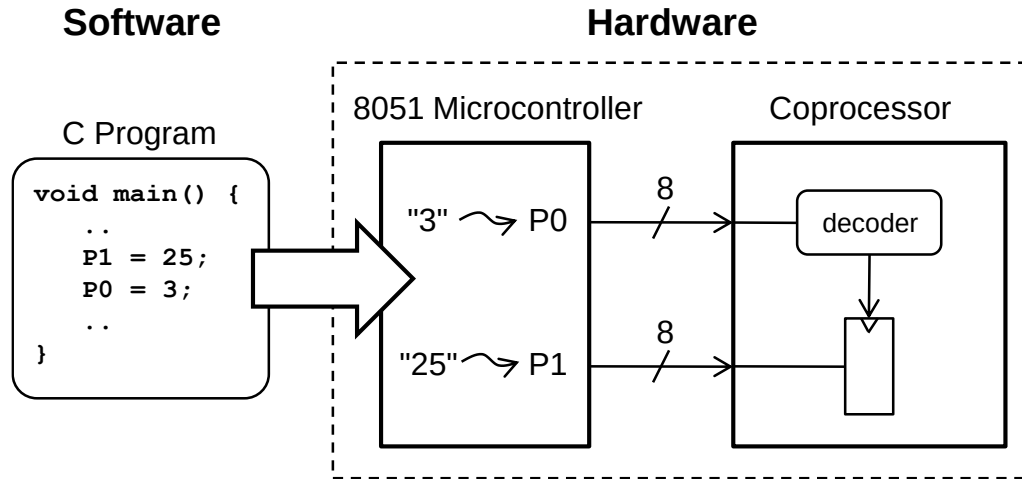


HW/SW Codesign

- ASIP (Application-Specific Instruction-set Processor)
 - Customizable instruction set
 - **Hardware:** can be extended, encapsulated as new instructions
- CELL processor



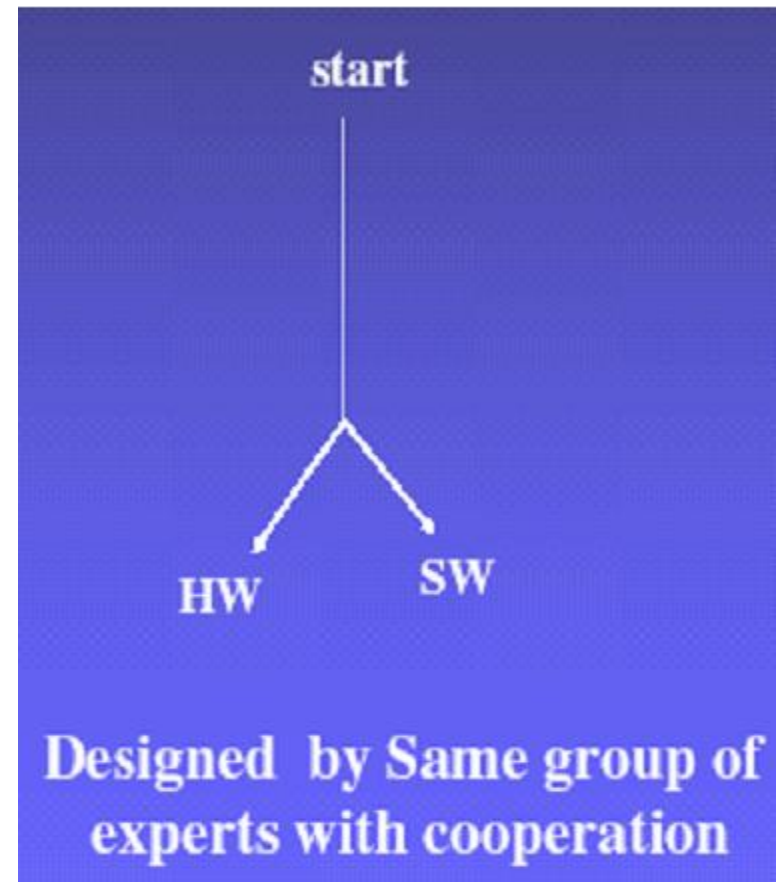
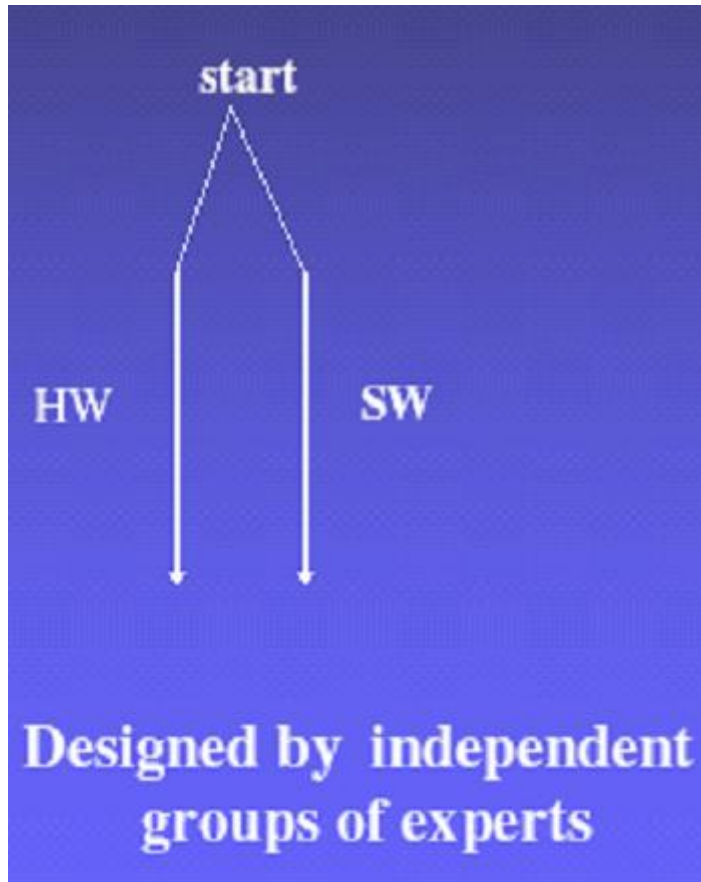
HW/SW Codesign



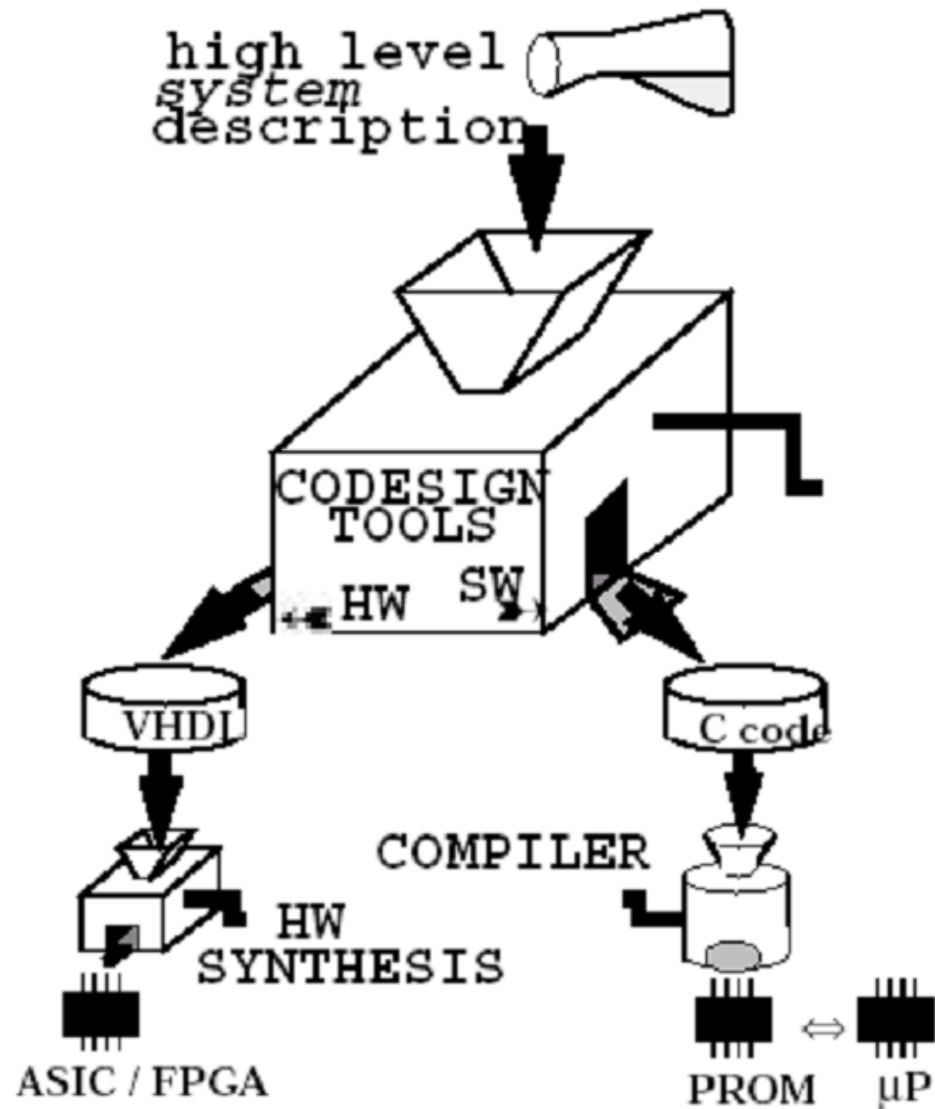
Hardware/Software Codesign is the partitioning and design of an application in terms of fixed and flexible components

HW/SW Codesign

Traditional design flow Concurrent (Codesign) flow



HW/SW Codesign



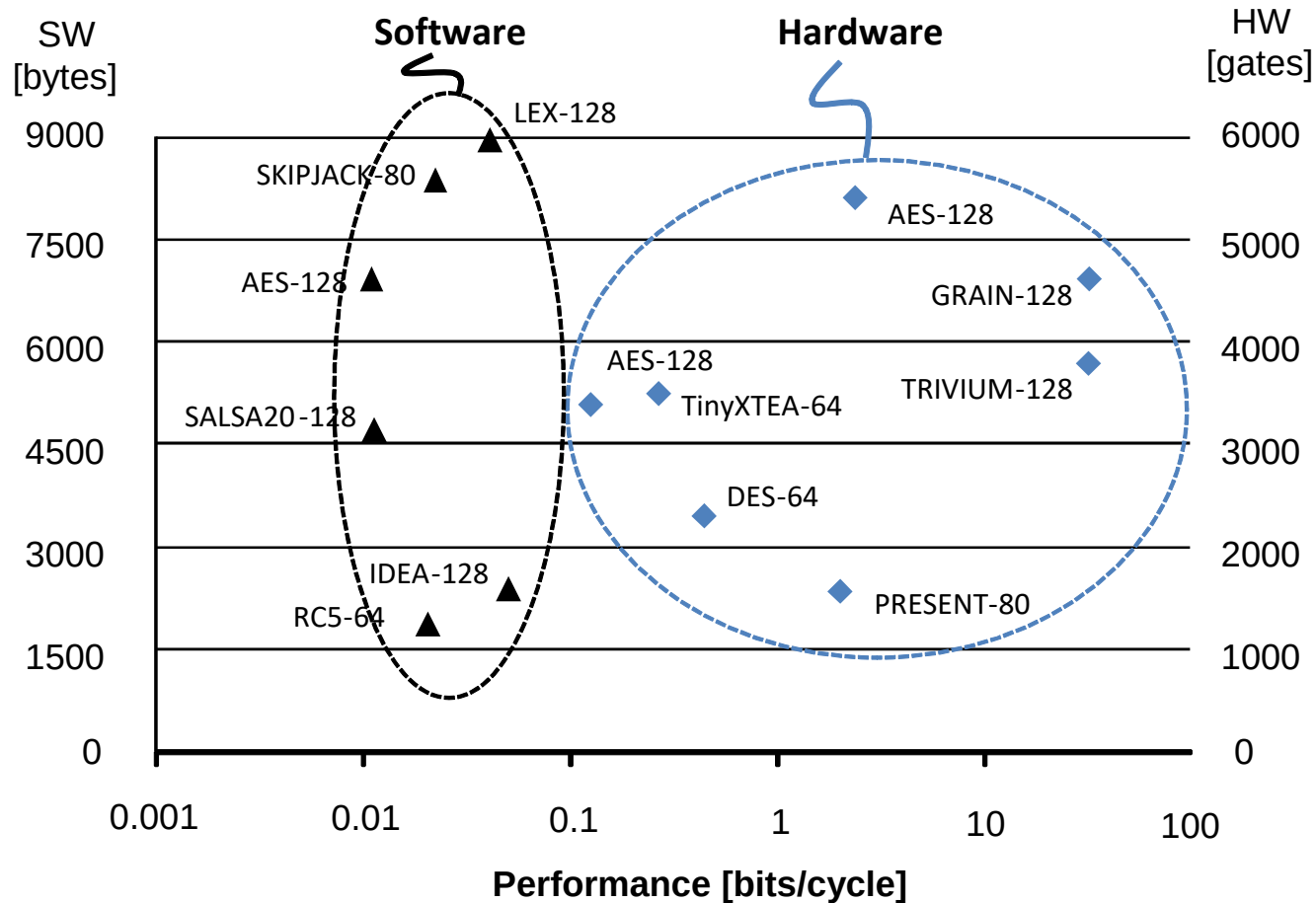
Energy Efficiency

- **Software:**
 - Easy and flexible
 - Software compilers are fast,
 - There are many libraries available
 - Development systems (personal computers and laptops) are plentiful and cheap.
- **Hardware architecture:** implementation, the RISC processor



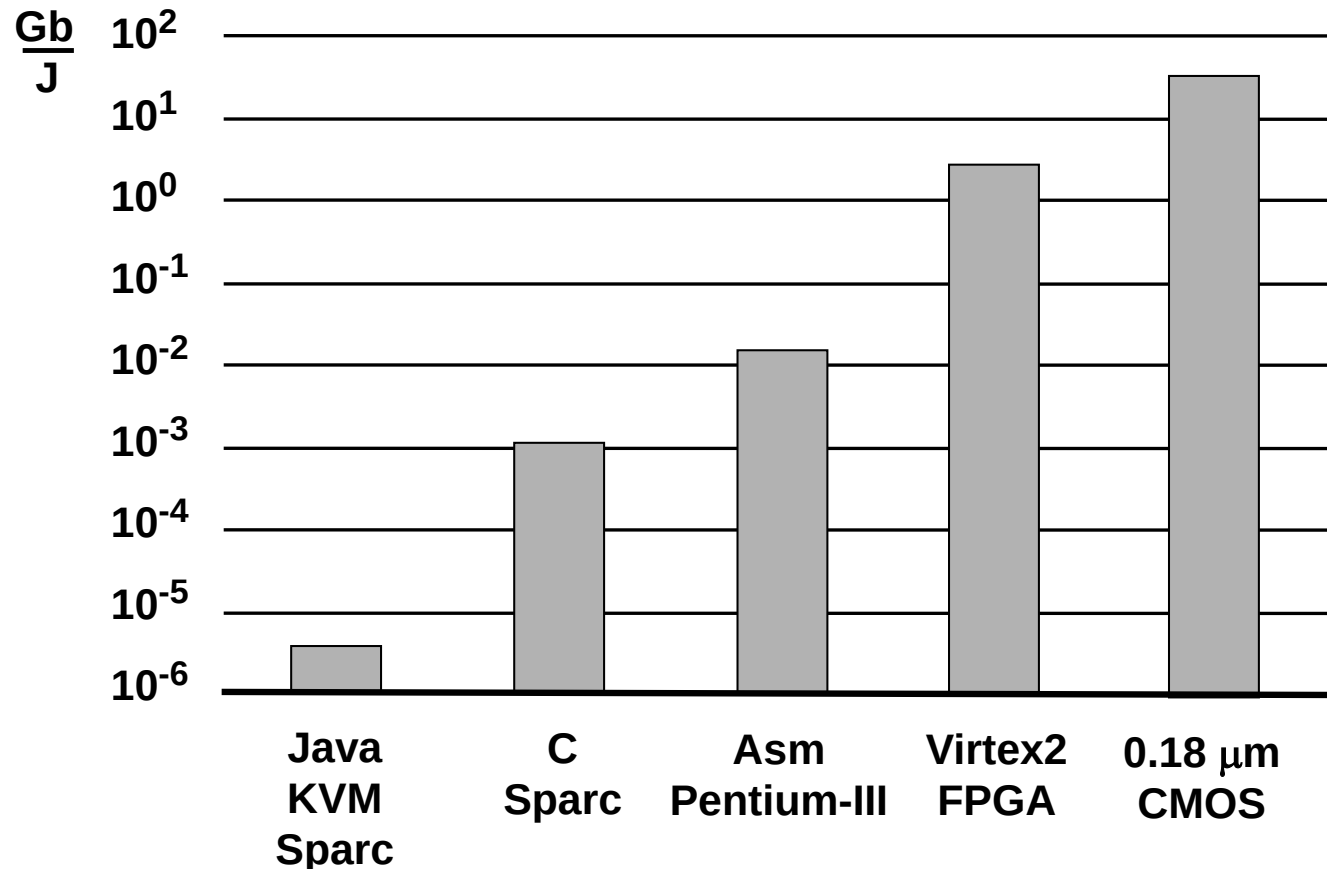
Energy Efficiency

- **Performance:** the amount of work done per unit of time

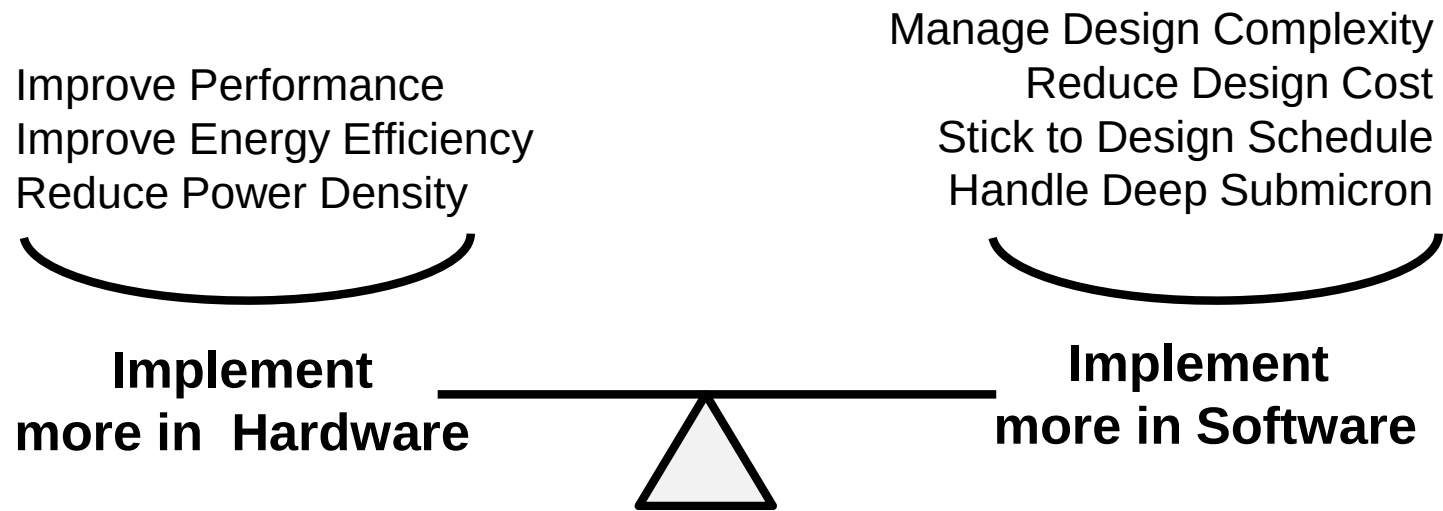


Energy Efficiency

- **Energy Efficiency:** the amount of useful work done per unit of energy



Energy Efficiency



Motivations

- Increasing capacity of field programmable devices
 - some devices even able to be reprogrammed on-the-fly (FPGAs, CPLDs...)
- Efficient C compilers for embedded processors
- Hardware synthesis capabilities

Motivations

- ❑ Improves design quality, design cycle time, and cost
 - ❑ Reduces integration and test time
- ❑ Supports growing complexity of embedded systems
- ❑ Takes advantage of advances in tools and technologies
 - ❑ Processor cores
 - ❑ High-level hardware synthesis capabilities
 - ❑ ASIC development

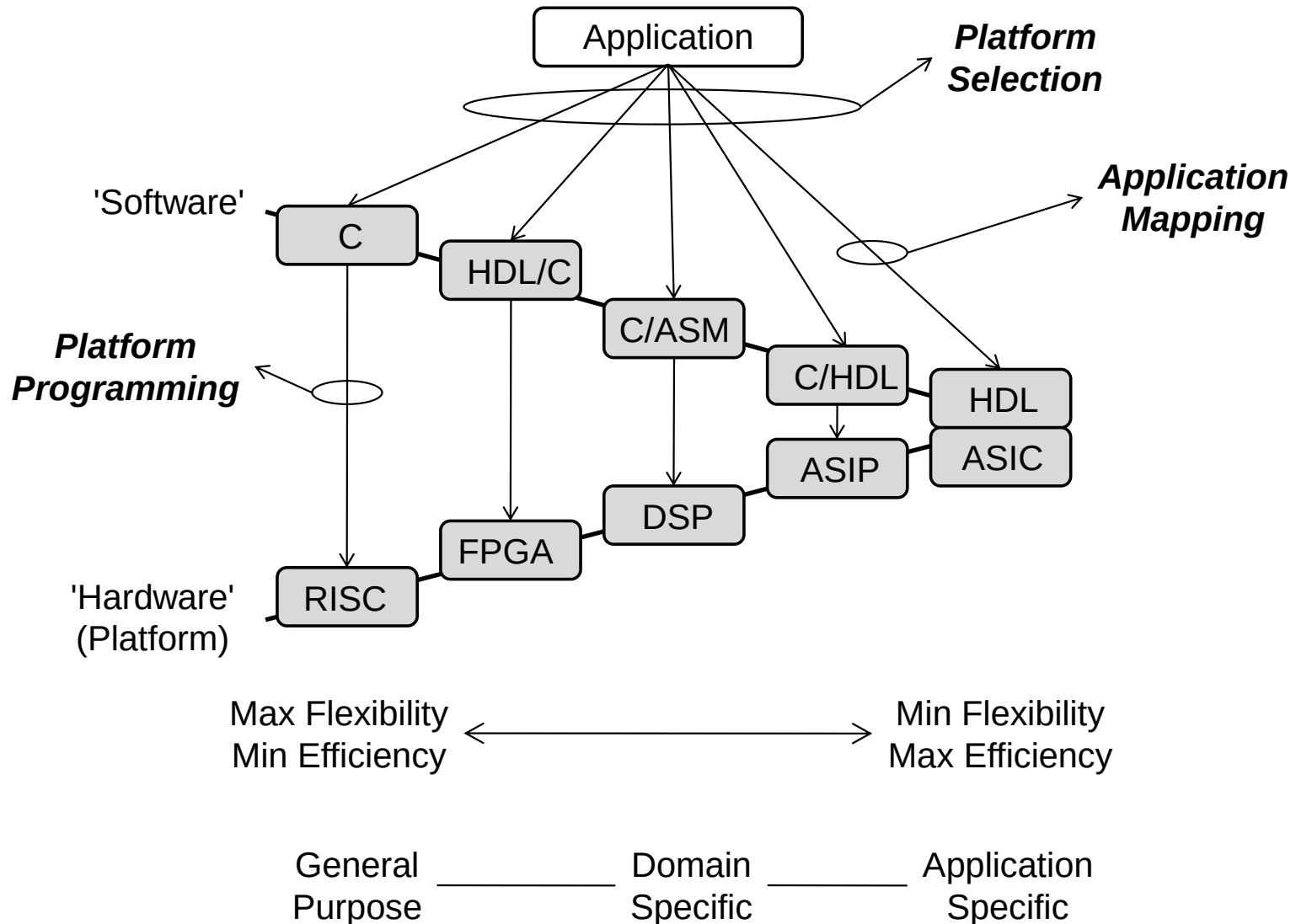
Steps in CD

HW-SW system involves

- **specification**
- **modeling**
- design space exploration and partitioning
- synthesis and optimization
- **validation**
- **implementation**

- A hardware/software ***partitioning*** represents a physical partition of system functionality into application-specific hardware and software.
- ***Scheduling*** is to assign an execution start time to each task in a set, where tasks are linked by some relations.

Platform Design Space



Research areas in CD

- ❑ Languages
- ❑ Architectural exploration tools
- ❑ Algorithms for partitioning
- ❑ Scheduling
- ❑ SW, HW and interface Synthesis
- ❑ Verification and Testing

References

- [1] Patrick R. Schaumont, A Practical Introduction to Hardware/Software Codesign, Springer Science & Business Media, 2nd Edition, 2010
- [2] Jørgen Staunstrup, Wayne Wolf, Hardware/Software Co-Design: Principles and Practice, Springer Science & Business Media, 1997
- [3] Dr. Sudarshan, Hardware Software Codesign, School of Engineering, Bangalore, 2012